



SPRINT 72

72nd BPSC Prelims Batch

Chemistry English Notes

1-Atomic Theory

Atomic theory explains the structure, behaviour, and interactions of matter at the atomic level. It states that all matter is made of tiny particles called atoms, the basic building blocks of matter. The theory evolved from ancient ideas of indivisible particles to modern atomic models through contributions of scientists like John Dalton, J. J. Thomson, Ernest Rutherford, Niels Bohr, and Erwin Schrödinger. Atomic theory forms the basis of chemistry, physics, material science, and engineering.

Dalton's Atomic Theory (1808)

In 1808, **John Dalton** proposed the first scientific atomic theory, known as **Dalton's Atomic Theory**, which identified atoms as the fundamental and smallest particles of matter. His theory successfully explained the laws of conservation of mass, constant composition, and multiple proportions.

- **Law of Conservation of Mass** (Antoine Lavoisier, 1789): Mass can neither be created nor destroyed during a chemical reaction.
- **Law of Constant Composition/Definite Proportions** (Joseph Proust): Elements in a chemical compound are always present in a fixed proportion by mass.
- **Law of Multiple Proportions** (Dalton, 1803): When two elements form more than one compound, the masses of one element combining with a fixed mass of the other are in simple whole-number ratios. For **example**, carbon and oxygen form **carbon monoxide (CO)** and **carbon dioxide (CO₂)**. For a fixed mass of carbon, the mass of oxygen combining with it is in the simple ratio of **1:2**.

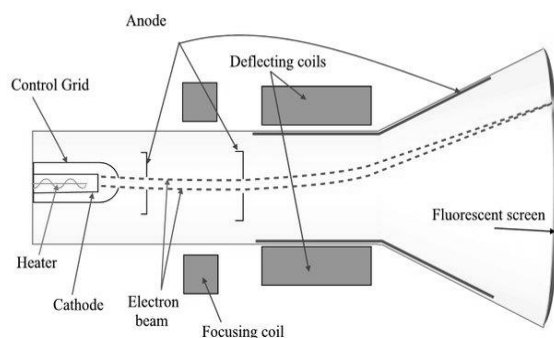
BCW Fact: The concept of the divisibility of matter was introduced in ancient India around **500 BC** by philosopher **Maharishi Kanad**. He proposed that continuous division of matter (*padarth*) would ultimately produce indivisible particles called **Parmanu (atoms)**. This idea was explained in his book **Vaisheshika Sutra**.

Atoms

Atoms are the **basic building blocks of matter**. Every substance in the universe, except energy, is made of atoms. An atom is further divisible and consists of three fundamental subatomic particles: **electron, proton, and neutron**.

Electron

- Electrons are **negatively charged particles** that revolve around the nucleus in fixed energy orbits.
- In cathode ray experiments, high voltage applied across electrodes produced rays moving from the cathode to the anode, called **cathode rays**. When these rays struck a **zinc sulphide coating**, a bright spot on the coating, confirming the presence of negatively charged sub atomic particles.
- The behaviour of cathode rays showed that they are streams of negatively charged particles called **electrons**.
- In 1897, J. J. Thomson measured the **charge-to-mass ratio (e/m)** of electrons using a cathode ray tube.
- Relative charge: **-1** and electron symbol: **'e'**.



- **Absolute charge:** $-1.60 \times 10^{-19} \text{C}$

BCW Fact: J. J. Thomson received the **Nobel Prize in Physics (1906)** for his discovery of the electron and his work on the conduction of electricity by gases. The term “**electron**” was coined by **G. J. Stoney** in 1891.

Proton

- These are **positively charged particles** present in the nucleus. The **smallest and lightest** positive ion obtained from hydrogen was named the proton.
- **Ernest Rutherford** established the proton as a fundamental particle in 1919.
- Relative charge: **+1** and proton symbol: '**p**'.
- **Absolute charge:** $1.60 \times 10^{-19} \text{C}$

Neutron

- Neutrons are **electrically neutral particles** present in the nucleus.
- In **1932**, James Chadwick discovered neutrons by bombarding **beryllium** with **α -particles** from the natural radioactive decay of polonium. The radiation showed **high penetrating power** no charge, confirming the existence of the **neutron**.
- When **electrically neutral particles (no electrical charge)** having a *mass slightly greater than that of protons* were emitted, it was named the **neutron**, establishing the existence of a **neutral subatomic particle** in the atom.
- **0** and '**n**' are relative charge and neutron symbol.
- **Absolute charge:** 0

BCW Fact: Atoms are **electrically neutral** because they contain equal numbers of **protons** and **electrons**. Protons and neutrons form the **nucleus**, while electrons occupy a surrounding **electron cloud**. The mass of an electron is about **1/2000th** that of a proton or neutron, so its contribution to atomic mass is virtually negligible. Atomic mass is measured in **atomic mass units (amu)**, where **1 amu = $1.66 \times 10^{-24} \text{g}$** .

Atomic Models

An atomic model is a representation of an atom showing its structure and arrangement of subatomic particles. Atomic models evolved over time to improve understanding of atomic structure. The five major atomic models were proposed by **John Dalton, J.J. Thomson, Ernest Rutherford, Niels Bohr, and Erwin Schrödinger**.

1. Plum Pudding Model

- J. J. Thomson proposed the **first atomic model** in **1898**, stating that the atom is a **uniform positively charged sphere** (radius $\approx 10^{-10} \text{m}$) with **electrons embedded** in it to maintain electrostatic stability. It is called the **Plum Pudding Model** because the atom was visualised as a **positively charged sphere** with embedded **electrons**, like plums/seeds in a pudding or watermelon.
- The model was later replaced by Ernest Rutherford's atomic model, which introduced the concept of the **atomic nucleus**.

2. Rutherford's Atomic Model

- Ernest Rutherford proposed the **nuclear/planetary model** of the atom in **1911**. According to this model, an atom has a **small, dense, positively charged nucleus** at the centre, around which the negatively charged **electrons** revolve like planets around the Sun.

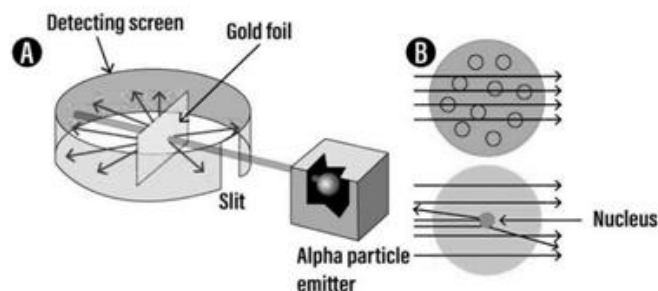
Gold Foil Experiment

- Rutherford bombarded a **thin gold foil** with **α -particles** and observed their deflection using a fluorescent **zinc sulphide screen**. The results of the experiment were unexpected:
- Most α -particles passed through the gold foil **without deflection**.
- A small fraction of α -particles was deflected through **small angles**.
- Very few α -particles (about **1 in 20,000**) bounced back, were deflected back nearly **180°**.

Conclusion:

From these observations, Rutherford concluded that:

- Most of the atom is **empty space**.
- Positive charge is concentrated in a **very small region**; and the positively charged part occupies very little space compared to the whole atom. The dense, positively charged central region was called the **nucleus**.
- Electrons revolve around the nucleus in **circular orbits** at very high speed, making the atom resemble the **solar system**. Electrons and nucleus are held together by **electrostatic force of attraction**.
- Electrostatic attraction between nucleus and electrons is balanced by **centrifugal force** (an apparent outward force on a mass when it is rotated).
- Radius of atom is about **10^5 times** the radius of nucleus.
- Volume of atom is about **10^{15} times** greater than that of nucleus.

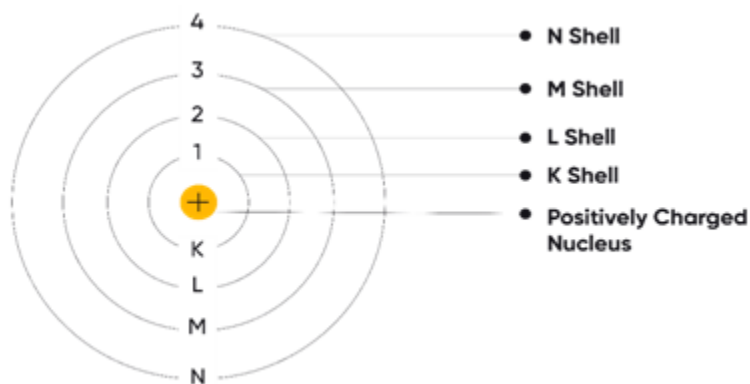
**Drawbacks:**

- Rutherford's model failed to explain the **stability of atoms**, as revolving electrons should continuously lose energy and fall into the nucleus.
- It could not explain the **electronic arrangement** around the nucleus.

3. Bohr Model

The **Bohr model of atom** was proposed by Niels Bohr in 1913 as a modification of Rutherford's model. Bohr's postulates:

- Electrons revolve around the positively charged nucleus in fixed circular paths called **orbits or shells** having **fixed energy**.
- Only certain **discrete orbits** of electron are allowed inside the atom.
- Electrons **do not radiate energy** while revolving in these orbits.
- Shells, known as energy levels, are represented as **K, L, M, N** or **$n = 1, 2, 3, 4, \dots$** , where **$n = 1$** has the lowest energy level.
- Electrons absorb energy to move to a higher level and release energy as **electromagnetic waves** while returning to a lower level.
- An atom is stable when the valence shell has **8 electrons (Law of Octet)**.
- Maximum electrons in n th shell is given by formula = **$2n^2$**



Shell And 'N' Value	Maximum Electrons Present In The Shell
K shell, n=1	$2 \times 1^2 = 2$
L shell, n=2	$2 \times 2^2 = 8$
M shell, n=3	$2 \times 3^2 = 18$
N shell, n=4	$2 \times 4^2 = 32$

Bohr explained atomic stability through electrons revolving in **fixed energy levels**. Electrons move from lower to higher energy levels by **absorbing energy** and from higher to lower levels by **releasing energy**.

BCW Fact: Antoine Lavoisier is known as the **father of modern chemistry**. He named the elements carbon, hydrogen and oxygen and discovered the role of oxygen in combustion and respiration.

Valency

- It is the **combining capacity** of an atom with atoms of the same or different elements to form molecules.
- An outermost shell with **8 electrons** is called an **octet**.
- Atoms gain stability by completing the octet in the outermost shell through **gaining, losing, or sharing electrons**.
- The number of electrons gained, lost, or shared is called the **valency** of the element.

Atomic Number (Z)

- It is the **total number of protons** present in the nucleus of an atom and is denoted by **Z**.
- All atoms of an element have the same atomic number.
- Example: Carbon $\rightarrow Z = 6$.

Atomic number (Z) = number of protons = number of electrons

Mass Number (A)

- It is the **sum of protons and neutrons** present in the nucleus of an atom and is denoted by **A**.
 $A = p + n$
- The mass of an atom is mainly due to the **nucleus**.
- Protons and neutrons together are called **nucleons**.

Mass number = number of protons + number of neutrons

Isotopes

Isotopes are atoms of the **same element** having the **same atomic number** but **different mass numbers** due to different numbers of neutrons.

- They have similar **chemical properties** but different **physical properties**.
- Many elements exist as a mixture of isotopes, and each isotope is a **pure substance**.

Examples:

- Hydrogen atom has three isotopes — **Protium** (${}^1_1\text{H}$) **Deuterium** (${}^2_1\text{H}$ or **D**), and **Tritium** (${}^3_1\text{H}$ or **T**) All have atomic number **1**, but mass numbers **1, 2, and 3** respectively.
- Deuterium reacts with oxygen to form **D₂O (heavy water)**.
- Carbon isotopes are ${}^{12}_6\text{C}$, ${}^{13}_6\text{C}$, ${}^{14}_6\text{C}$
- Carbon-14** (${}^{14}_6\text{C}$ or **radiocarbon**) is a radioactive isotope of carbon containing **6 protons and 8 neutrons**. It is used in **radiocarbon dating** to determine the age of archaeological, geological, and hydrogeological samples.

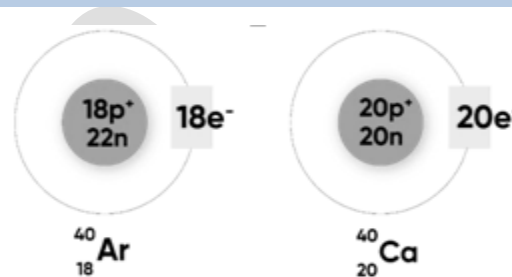
BCW Fact: Heavy Water (D_2O) or deuterium oxide is made up of two atoms of deuterium and one atom of oxygen. Heavy water displays similar physical and chemical properties but differs in nuclear properties when compared to ordinary water which makes it an extremely efficient material for use as **moderator** and **coolant** in nuclear reactors.

Applications of Isotopes:

- **Iodine isotope** → treatment of **goitre**.
- **Uranium isotope** → fuel in **nuclear reactors**.
- **Cobalt-60 isotope** → treatment of **cancer/leukaemia**.

Isobars

- Atoms of different elements having **different atomic numbers** but the **same mass number** are called **isobars**. Thus, they have the same number of **nucleons**.
- **Example:** Calcium (${}^{40}_{20}\text{Ca}$) and Argon (${}^{40}_{18}\text{Ar}$) have different atomic numbers (20 and 18 respectively), but the same mass number **40**. While the number of electrons differs in these atoms, their **mass number remains the same**.



Isotones

- Atoms having the **same number of neutrons** but different numbers of protons are called **isotones**.
- **Example:** ${}^{14}_6\text{C}$ and ${}^{15}_7\text{N}$ both contain **8 neutrons**, but have different proton numbers **6 and 7** respectively.

Isoelectronic

- Atoms or ions having the **same number of electrons** are called **isoelectronic species**.
- **Examples:** O^{2-} , F^- , Ne

Isosters

- Molecules or polyatomic ions having the **same number of atoms** and **same number of electrons** are called **isosters**.
- **Examples:** N_2 , CO , CN^-

BCW Fact: Ernest Rutherford is known as the “**Father of Nuclear Physics**”. However, he received **Nobel Prize for “Chemistry” in 1908** for his investigations into the disintegration of elements and the chemistry of radioactive substances.

Quantum Number

The set of four integers required to define an electron completely in an atom are called quantum numbers. They are:

- Principle quantum number (n): It indicates the energy level and size of the orbital.
- Azimuthal quantum number (l): It determines the shape of the orbital (subshell).
- Magnetic quantum number (m): Specifies the orientation of the orbital.
- Spin quantum number (s): Specifies the direction of the electron's spin.

Pauli's Exclusion Principle

- The number of electrons to be filled in various orbitals is restricted by the exclusion principle, given by the Austrian scientist **Wolfgang Pauli (1926)**. According to this principle:
- All the **four quantum numbers** for **any two electrons** in an atom *cannot be identical*.

Aufbau Principle

- The electrons are filled in various orbitals in **order of their increasing energies**.
- In other words, electrons first occupy the lowest energy orbital available to them and enter into higher energy orbitals only after the lower energy orbitals are filled.

2- Matter, Solution And Colloids

Matter, solutions, and colloids form the basis of understanding the physical and chemical nature of substances. **Matter** is anything that has mass and occupies space. It mainly exists as **solid, liquid, and gas**, while science also recognizes **two more states**— **plasma** and **Bose–Einstein condensate (BEC)**. **Solutions** are homogeneous mixtures showing properties like boiling point elevation and freezing point depression. **Colloids** are heterogeneous mixtures with particle sizes of **1–1000 nm** that exhibit effects like the **Tyndall effect**.

What Is Matter?

Matter is anything that has **mass and occupies space**. All everyday objects are made of atoms and subatomic particles.

Ancient Indian philosophers described matter through the “**Panch Tattvas**”—air, earth, fire, sky, and water. Matter is composed of tiny particles having different properties.

Characteristics Of Particles Of Matter

These below mentioned properties determine the state of matter.

a. Particles have space between them

- Matter contains **inter-particle spaces**, allowing particles of one substance to mix with another.
- **Example:** perfume mixes in air, and sugar dissolves in water.

b. Particles possess continuous movement

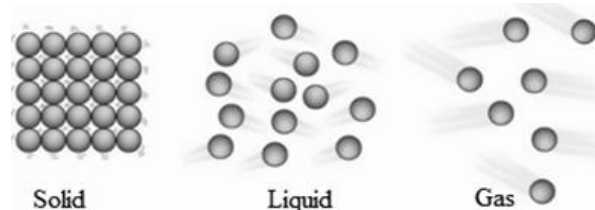
- Particles of matter continuously move because they possess **kinetic energy**. Their kinetic energy increases with temperature.
 - The particles of matter **intermix themselves** with each other by going into the inter spaces. Intermixing of particles is called **diffusion**.
 - **Diffusion increases** on heating.
 - Diffusion is **faster in gases** and slowest in solids because gases have maximum inter-particle space and kinetic energy.

c. Particles attract each other

- Particles are held together by **intermolecular force of attraction**. Solids have the strongest attractive forces, minimal inter-particle space, and low kinetic energy, making them rigid and compact.

States Of Matter

Matter exists in different states or phases. The three common states are **solid, liquid, and gas**, while other states include plasma, and Bose–Einstein condensate (BEC). These states arise due to differences in the properties and arrangement of particles. For example, Water exists as **ice (solid), water (liquid), and steam (gas)**.



1. Solid

- Solids have **fixed shape and volume** with negligible compressibility.
- Solids maintain shape under force.
- Solids may break under force but they are rigid and difficult to deform.

NOTE: A rubber band is considered a solid because it is an elastic solid and **it changes shape when a force is applied**. However, the rubber band **regains its original shape when the force is removed**, which is a characteristic of solids. However, if excessive force is applied, the rubber band breaks, demonstrating the elastic limit of solids.

2. Liquid

- Liquids have **fixed volume but no fixed shape**.
- They flow and take the shape of the container.
- Dissolved oxygen in water allows aquatic animals to breathe. So, it is observed that solids, liquids and gases can diffuse into liquids.

3. Gas

- Gases have **no fixed shape or volume**. They are highly compressible as compared to solids and liquids.

Particulars	Solid	Liquid	Gas
Characteristic	Rigid, Fixed shape & Volume	Fluid, no fixed shape but fixed volume	no fixed shape & volume
Forces of attraction between the particles	Maximum	Intermediate	Minimum
Spaces between particles	Minimum	Intermediate	Maximum
Kinetic energy of the particles	Minimum	Intermediate	Maximum

4. Plasma

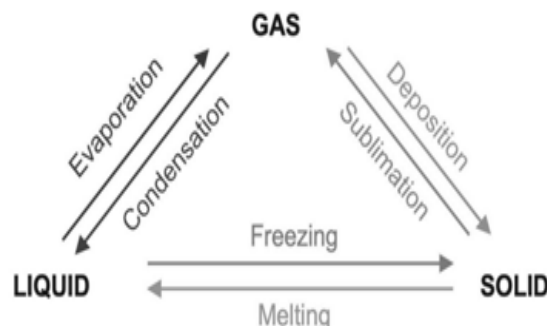
- **Plasma** is the fourth state of matter after solid, liquid, and gas. It consists of **ionized gases** containing *electrons, ions, neutral atoms, and free radicals*.
- In this state, gases become extremely hot, causing atoms to lose or gain electrons and become electrically charged.
- Plasma can conduct electricity and is influenced by magnetic fields. It is the most abundant state of matter in the universe, and stars are primarily made of plasma.

5. Bose–Einstein Condensate (BEC)

- BEC is the **fifth state of matter**, discovered mainly through the work of Satyendra Nath Bose. It is a **supercooled state** formed when particles called **bosons** are cooled to **near absolute zero temperature**. The term “**boson**” was given by Paul Dirac to honour Bose’s contribution.

Change In States Of Matter

A **change of state** is a **physical change** in matter that is usually **reversible** and does not alter its chemical composition. Processes like **melting**, **freezing**, **evaporation**, **condensation**, and **sublimation** are examples of changes of state. Matter can exist in different states; for example, water occurs as **ice (solid)**, **water (liquid)**, and **water vapour (gas)**. These states are interconvertible by changing **temperature or pressure**.



Temperature

- Temperature is the **degree of hotness or coldness** of a body. It refers to the degree of heat present in a matter.
- Increasing temperature (or thermal energy is added to matter) increases the **kinetic energy** of particles, helping them overcome intermolecular attraction. Cooling has the reverse effect.

Solids → Liquids → Gas (with increase in temperature), K.E increases

A. Solid ⇌ Liquid

1. **Melting:**

- Melting is the conversion of state of a substance from **solid into liquid**.
- In solids, particles are tightly packed due to strong intermolecular forces. On heating, their **kinetic energy increases**, allowing particles to overcome the force of attraction and convert the **solid into a liquid**.
- Melting point:** The minimum temperature at which solid converts to liquid.
- Latent heat of fusion:** The amount of heat energy required to convert 1 kg solid into liquid at its melting point.

2. **Freezing:**

- Freezing is the conversion of state of a substance from **liquid into solid**.
- As temperature decreases, the **kinetic energy of particles reduces**, causing them to come closer due to stronger intermolecular attraction. Below the **freezing point**, the liquid changes into a **solid**.
- Freezing point:** The temperature at which liquid changes into solid at atmospheric pressure.
- Latent heat of freezing:** The amount of heat energy released when liquid solidifies into solid.

B. Liquid ⇌ Gas

1. **Vapourisation:**

- Conversion of state of a substance from **liquid into gas** is called vapourisation.
- In liquids, particles are held together by **moderate intermolecular forces**. On heating, their **kinetic energy increases**, enabling them to overcome the force of attraction and change from **liquid to gas**.
- Boiling point:** The temperature at which the **vapour pressure of a liquid equals atmospheric pressure**. Under this condition, added heat converts the liquid into vapour **without increasing the temperature**.
- Latent heat of vapourisation:** The heat energy released when 1 kg of a liquid changes into a gas at atmospheric pressure at its boiling point.

2. **Condensation:**

- The process in which state of a substance **changes from gas into a liquid**, and is the reverse of vapourisation.

- When vapour particles are cooled below the **boiling point**, their **kinetic energy decreases** and intermolecular attraction becomes dominant, causing the gas to convert into a **liquid**.
- **Condensation point**: The temperature at which a gas changes into a liquid. In most cases, the boiling point and condensation point are the same.
- **Latent heat of condensation**: The heat released when **1 kg of vapour changes into liquid** at its condensation point without temperature change.

C. $\text{Gas} \rightleftharpoons \text{Solid}$

1. Sublimation

- Sublimation is direct conversion of state of a substance from **solid into gas** without passing through liquid state.
- It is an **endothermic process** that occurs at temperatures and pressures **below the triple point** of a substance.

3. Deposition

- The process in which a **gas changes directly into a solid** without passing through the liquid state. It is the **reverse of sublimation** and is also called **de-sublimation**.
- **Example**: iodine vapours forming iodine crystals on cooling.

BCW Fact: The **triple point** is the temperature and pressure where solid, liquid, and vapour coexist in equilibrium.

Anomalous Behaviour of Water

Normally, substances **contract on cooling**, but water behaves differently below **4°C**. Water contracts up to **4°C**, but on further cooling it **expands**, a phenomenon called the **anomalous behaviour of water**.

- Water has **maximum density at 4°C**.
- Below 4°C, water expands due to **hydrogen bonding**, which increases volume and decreases density.

Effect Of Pressure On Solution (Henry's Law)

- **Henry's law** states that at constant temperature, the solubility of a gas in a liquid is directly proportional to the **partial pressure** of the gas above the liquid.
- When solubility is measured using the **mole fraction of the gas** in the solution, Henry's law can be expressed as follows:
 - **The mole fraction or mass of a gas** dissolved in a solvent is directly proportional to the partial pressure of the gas.

Applications of Henry's Law

- Soft drinks and soda water are **sealed under high pressure** to increase the solubility of CO_2 .
- Scuba divers breathe air under high pressure underwater, which increases the solubility of gases in the blood. During ascent, pressure decreases, reducing nitrogen solubility and causing nitrogen bubbles to form in the blood, leading to a dangerous condition called the **bends**. Hence, helium is mixed with oxygen in diving cylinders.
- At higher altitudes, the **pressure of oxygen** is lower than at ground level. This reduces the **oxygen concentration in blood and tissues** of mountaineers and people living at high altitudes. Low blood oxygen makes climbers **weak, fatigued, and mentally confused**. Apparently, develops the symptoms of a condition called anoxia.

Evaporation

Evaporation is the slow conversion of liquid into vapour at any temperature. It is a **surface phenomenon** and produces a **cooling effect**.

Factors Affecting Evaporation

1. **Surface Area:** Larger surface area increases evaporation.
2. **Temperature:** Higher temperature increases kinetic energy and evaporation rate.
3. **Humidity:** Lower humidity increases evaporation, while high humidity decreases it.
4. **Wind Speed:** Higher wind speed removes vapour particles, increasing evaporation.

Applications of Evaporation

- Sweating cools the body through **evaporative cooling**. The degree of cooling depends on the evaporation rate and the heat of vaporization.
- After a hot sunny day, people sprinkle water on roofs or open grounds because the **high latent heat of vaporization of water** cools the hot surface.
- Water stored in an **earthen pot (matka)** becomes cool due to evaporation through its tiny pores. As water evaporates from the pot's surface, it absorbs heat from the water inside, making it cooler than the surroundings.

BCW Fact: **Vapour Pressure** is a measure of tendency of a substance to change into gaseous state.

Gas Laws

Gas is a state of matter whose molecules or atoms are in constant random motion. Gas laws describe the behaviour of gases by establishing relationships among the following variables: **volume** (space occupied by the gas), **pressure** exerted by the gas on the walls of the container, **Absolute temperature** of the gas, and amount of gaseous substance (number of moles).

- **Boyle's law:** The volume of a gas is inversely proportional to its pressure at a constant temperature and mass.
- **Charles's law** - The volume of a gas is directly proportional to its absolute temperature at a constant pressure and mass.
- **Gay-Lussac's law** - The pressure of a gas is directly proportional to its temperature at a constant volume and mass.
- **Avogadro's law** - The volume of a gas is directly proportional to the number of moles of gas at a constant temperature and pressure.
- **The ideal gas law is, $PV = nRT$**

The gas which follows this equation is called an *Ideal gas equation* and it can be derived by combining Boyle's Law, Charles' Law, and Avogadro's Law.

where P is the pressure of the gas, V is the volume of the gas, n is the number of moles, **R is the universal gas constant** = $8.3145 \text{ J.mol}^{-1}\text{K}^{-1}$, T is the temperature of the gas, and N is Avogadro's number, **$N_A = 6.0221 \times 10^{23}$**

Kinetic Molecular Theory Of Gases

1. Gases consist of large number of tiny particles called **molecules**.
2. The size of gas molecules is extremely small compared to the total volume occupied by the gas; hence, the **volume of individual molecules is negligible**.
3. Under ordinary temperature and pressure, gas particles **do not experience significant intermolecular attractive forces**.
4. Gas particles are in **constant random motion**.
5. Gas particles **move in straight lines in all possible directions**.
6. Collisions between gas molecules are **perfectly elastic**; therefore, the total kinetic energy remains constant before and after collision.
7. The **average kinetic energy** of gas particles is **directly proportional to the absolute temperature** of the gas.

Physical And Chemical Changes

Matter undergoes changes that are classified as either **physical** or **chemical** changes.

Physical change

- A **physical change** is usually **reversible and does not alter the chemical composition** of a substance. For example, an ice cube melts into water and can be frozen back into ice. In such changes, the material remains structurally the same before and after the change.
- Physical changes mainly involve alterations in **shape, size, texture, temperature, or state of matter**.
- **Examples:** melting of ice, vapourisation of water, crushing a can, dissolving sugar in water, mixing sand and water, breaking glass, and crumpling paper bags.

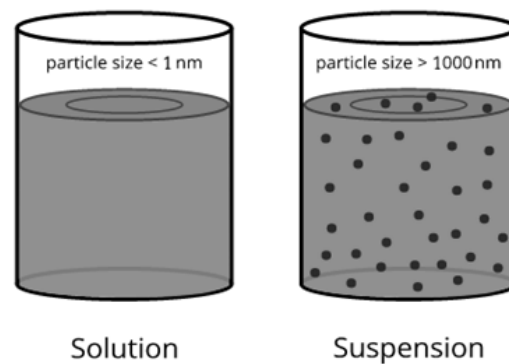
Chemical change

- A **chemical change** involves a **change in the chemical composition** of a substance due to the breaking and formation of chemical bonds.
- During a chemical reaction, atoms rearrange to form new substances that cannot be easily converted back to their original form.
- Generally **irreversible** and are often accompanied by energy changes, resulting in exothermic or endothermic reactions.
- **Examples:** cooking food, formation of curd, digestion of food, rusting of iron, ripening of fruits, setting of cement, and burning of coal.

BCW Fact: Burning of candle is an *example of both physical and chemical change*. Melting of wax is physical change, while **formation of carbon dioxide during burning** is a chemical change.

Solution

- A **solution** is a homogeneous mixture of two or more substances. The component present in a larger amount is called the **solvent**, while the component present in a smaller amount is called the **solute**.
- A homogeneous mixture has **uniform composition and properties** throughout the mixture. The particle size in a true solution is **less than 1 nm**, making the particles invisible to the naked eye.
- Due to their extremely small size, the **particles do not scatter light; therefore, the path of light is not visible**. Solutions are stable mixtures because the solutes are **inseparable** from the solution and **do not precipitate**.



Key Related Terms:

- Dilute Solution** – A solution containing a small amount of solute compared to the solvent.
- Concentrated Solution** – A solution containing a relatively large amount of solute.
- Saturated Solution** – A solution in which no more solute can dissolve at a given temperature.
- Unsaturated Solution** – A solution that contains less solute than the maximum amount it can dissolve at a given temperature.
- Solubility** – The maximum amount of solute that can dissolve in a solvent at a given temperature to form a saturated solution.
- Volatile Compounds** – Substances that evaporate easily due to high vapour pressure, low boiling point, and weak intermolecular forces.
- Examples:** acetone, hexane, and chloroform.
- Non-Volatile Compounds** – Substances that do not evaporate easily because of low vapour pressure, high boiling point, and strong intermolecular forces.
- Examples:** glycerine, water, and oil.

Types of Solutions

Type Of Solution	Solute	Solvent	Common Examples
Gaseous Solutions	Gas	Gas	Mixture of oxygen and nitrogen gases
	Liquid	Gas	Chloroform mixed with nitrogen gas
	Solid	Gas	Camphor in nitrogen gas
Liquid Solutions	Gas	Liquid	Oxygen dissolved in water
	Liquid	Liquid	Ethanol dissolved in water
	Solid	Liquid	Glucose dissolved in water
Solid Solutions	Gas	Solid	Solution of hydrogen in palladium
	Liquid	Solid	Amalgam of mercury with sodium
	Solid	Solid	Copper dissolved in gold

Concentration Of Solutions

The concentration of a solution refers to the amount of solute dissolved in a given quantity of solvent or solution. It may also be defined as the amount of solute present per unit mass or per unit volume of the solvent/solution.

Methods of Expressing Concentration**1. Mass Percentage (Weight Percentage)**

- The mass of the solute divided by the mass of the solution, multiplied by 100.
Mass Percentage of Solute = $(\text{Mass of Solute} / \text{Mass of Solution}) \times 100$
- Example:** A 5% sugar solution contains 5 g sugar in 100 g solution.

2. Molarity (M)

Molarity is the number of moles of solute dissolved in one litre of solution.

- Molarity (M) = No. of Moles of Solute/Volume of Solution in Litres.
- Example:** A 1 molar solution contains 1 mole of solute in 1 litre of solution.

3. Molality (m)

Molality is the number of moles of solute dissolved in 1 kg of solvent.

- Molality (m) = No. of Moles of Solute/Mass of Solvent in Kilograms.
- Example:** A 1 molal solution contains 1 mole of solute in 1 kg of solvent.

4. Mole Fraction

Mole fraction is the ratio of moles of one component to the total moles of all components.

- Mole Fraction of Solute = Moles of Solute/Total Moles of Solution
- Example:** If a solution contains 2 moles of solute and 4 moles of solvent, the mole fraction of solute is 0.33.

5. Volume Percentage

Volume percentage is used when both solute and solvent are liquids. It is the volume of solute divided by the total volume of the solution, multiplied by 100.

- Volume Percentage of Solute = $(\text{Volume of Solute} / \text{Total Volume of Solution}) \times 100$
- Example:** A 10% alcohol solution means that 10 ml of alcohol is dissolved in 100 ml of solution.

6. Mass by Volume Percentage

- Mass by volume percentage (often abbreviated as % m/v) expresses the mass of the solute in grams per 100 ml of solution.
- Commonly used in medicine and pharmacy.

7. Parts Per Million (PPM)

- PPM refers to the number of parts of solute per million parts of solution.
- It used to express the concentration of a solute in a solution when the solute is present in very small, trace quantities such as pollutants in water.
- Formula:** $\text{PPM} = (\text{Mass of solute (in grams)} / \text{Mass of Solution (in grams)}) \times 10^6$
- Example:** 1 PPM means 1 part of solute in 1 million parts of solution.
- Similar to percentages, concentration in parts per million (PPM) can also be expressed in terms of mass to mass, volume to volume, and mass to volume.

Difference Between Evaporation And Boiling

Evaporation	Boiling
-------------	---------

Evaporation is a natural phenomenon in which liquid form changes into the gaseous form.	Boiling is an artificial phenomenon where the liquid gets heated up and vaporized because of continuous supply of heat.
It occurs even at normal temperature and pressure .	It only occurs at boiling point , which is the temperature at which vapour pressure of liquid is equal to atmospheric pressure.
This is a surface phenomenon where surface molecules gain energy to escape from the surface in gaseous form.	Boiling is a bulk phenomenon where the entire mass of the liquid gets heated up so any molecule from bulk can gain energy to vaporize.
Evaporation is a slow process and followed by cooling.	Boiling is a very fast process and vapour formation is rapid.

Colligative Properties

Colligative properties are physical properties of solutions that change when a solute is added to a solvent. These properties depend on the number of solute particles present, not on the nature of the solute.

Relative Lowering of Vapour Pressure

- The **relative lowering of vapour pressure** is the ratio of the decrease in vapour pressure of a solution to the vapour pressure of the pure solvent at the same temperature.
- When a non-volatile solute is added to a volatile solvent, vapour pressure decreases because solute particles occupy surface sites, reducing the number of solvent molecules escaping into the vapour phase. Since evaporation is a surface phenomenon, this leads to lower vapour pressure.
- According to **Raoult's Law**, the vapour pressure of a solvent above a solution is equal to the vapour pressure of the pure solvent multiplied by the mole fraction of the solvent present in the solution.

Depression of Freezing Point

- Freezing point depression** is the lowering of the freezing point of a solvent when a solute is dissolved in it.
- The freezing point is the temperature at which the vapour pressure of the liquid phase becomes equal to that of the solid phase. When a non-volatile solute is added, the vapour pressure of the solvent decreases. Therefore, the solution reaches equilibrium with the solid phase at a lower temperature, causing a decrease in the freezing point.

BCW Fact: Ethylene glycol is added to water as an **antifreeze** in car engines. It lowers the freezing point of water, preventing freezing in winter and also acts as a coolant in summer.

Elevation of Boiling Point

- The elevation of the boiling point of a solution is the **difference in temperature between the boiling point of the solution and that of the pure solvent**.
- A liquid boils when its vapour pressure becomes equal to atmospheric pressure. Since the addition of a non-volatile solute lowers the vapour pressure of the solvent, a higher temperature is required for the solution to reach atmospheric pressure. As a result, the boiling point increases.
- For example**, the vapour pressure of an aqueous solution of **sugar is less than 1.013 bar at 373.15 K**. In order to make this solution boil, its vapour pressure must be increased to 1.013 bar by raising the

temperature above the boiling temperature of the pure solvent (water). Thus, the boiling point of a solution is always higher than that of the boiling point of the pure solvent in which the solution is prepared.

Osmotic Pressure

- The spontaneous flow of the solvent from a solution of lower concentration to a solution of higher concentration when the two are separated by a semipermeable membrane is known as **osmosis**.
- Osmotic pressure is directly proportional to the concentration of the solution.**
- The osmotic pressure of a solution is the excess pressure that must be applied to the solution in order to prevent osmosis, that is, to prevent the passage of solvent molecules through a semi-permeable membrane into the solution.
- Reverse osmosis (RO):** Used in water purification.

Colloids

- A **colloid** is a heterogeneous system in which one substance is **dispersed** (dispersed phase) as very fine particles in another substance called the **dispersion medium**.
Example: toothpaste, smoke, fog, milk.
- The main difference between a solution and a colloid is **particle size**. In a solution, particles are ions or small molecules, whereas in a colloid the dispersed phase consists of larger particles such as macromolecules (proteins, polymers) or aggregates of atoms, ions, or molecules.
- Colloidal particles are larger than simple molecules but small enough to remain suspended. Their diameter ranges from 10^{-9} m to 10^{-7} m.

Types of Colloids

Colloids are classified on the basis of:

- Physical state of dispersed phase and dispersion medium
- Nature of interaction between them
- Type of particles of dispersed phase

BCW Fact: An **emulsion** is a mixture of two or more immiscible liquids. It is a liquid colloidal system in which both the dispersion medium and the dispersed substance are in the liquid phase.

A. Based on physical state of dispersed phase and dispersion medium:

- Depending on whether the dispersed phase and dispersion medium are **solid, liquid, or gas**, **eight types of colloidal systems** are possible. A **gas mixed with another gas forms a homogeneous mixture**, so it is **not considered a colloidal system**.

Types of colloidal system

Dispersed Phase	Dispersion Medium	Type Of Colloid	Examples
Solid	Solid	Solid sol	Some coloured glasses and gem stones
Solid	Liquid	Sol	Paints, cell fluids
Solid	Gas	Aerosol	Smoke, dust

Liquid	Solid	Gel	Cheese, butter, jellies
Liquid	Liquid	Emulsion	Milk, hair cream
Liquid	Gas	Aerosol	Fog, mist, cloud, insecticide sprays
Gas	Solid	Solid sol	Pumice stone, foam rubber
Gas	Liquid	Foam	Froth, whipped cream, soap lather

- Many common commercial products and natural substances are colloids. For example, **whipped cream** is a **foam (gas dispersed in liquid)**, and **fire-fighting foams used in emergency aircraft landings are also colloids**.
- Most biological fluids are aqueous** where solids are dispersed in water. In cells, proteins and nucleic acids act as colloidal-sized particles dispersed in an aqueous solution of ions and small molecules.
- Among different types of colloids, the most common are **sols (solid in liquid)**, **gels (liquid in solid)**, and **emulsions (liquid in liquid)**.
- If the dispersion medium is water, the sol is called **hydrosol (or aquasol)**.

B. Based on Interaction with Medium

Based on interaction with medium colloids are of two types:

1. **Lyophilic (solvent-attracting):** strong attraction between dispersed phase and dispersion medium.

- Lyophilic sols** mean “liquid-loving” sols. They are formed by directly mixing substances like **gum, gelatin, starch, and rubber** with a suitable liquid (dispersion medium).
- If the dispersion medium is removed, the sol can be **re-formed by simple mixing**, so they are called **reversible sols**. These sols are also **highly stable and difficult to coagulate**.

2. **Lyophobic (solvent-hating)** weak attraction between dispersed phase and dispersion medium.

- Lyophobic sols** mean “liquid-hating” sols. Substances like **metals and their sulphides** do not form colloids by simple mixing with the dispersion medium and require special preparation methods.
- Are unstable and easily coagulated** by small amounts of electrolytes, heating, or shaking.
- Once precipitated, they **cannot be reconverted into sols by simple addition of the dispersion medium**, so they are called **irreversible sols**. They also require **stabilizing agents** for stability.

NOTE: If water is the medium:

- Lyophilic → **hydrophilic**
- Lyophobic → **hydrophobic**

C. Based on Type of Particles of Dispersed Phase

Colloids are classified into:

- Multimolecular colloids
- Macromolecular colloids
- Associated colloids (micelles)

1. Multimolecular Colloids

- On dissolution, a large number of atoms or smaller molecules of a substance aggregate to form particles in the colloidal size range (1–1000 nm). These are called **multimolecular colloids**.
- Example:** Gold sol (contains particles with many atoms), sulphur sol (aggregates of S_8 sulphur molecules).

2. Macromolecular Colloids

- Macromolecules, when dissolved in suitable solvents, form solutions in which the particle size lies in the colloidal range. These are called **macromolecular colloids**. They are stable and resemble true solutions.
- **Examples:** Natural Macromolecule—starch, cellulose, proteins, enzymes; Man-made Macromolecule—polythene, nylon, polystyrene, synthetic rubber.

3. Associated Colloids (Micelles)

- Some substances behave as strong electrolytes at low concentration but form colloidal aggregates at higher concentration. These aggregates are called **micelles or Associated Colloids**.
- Micelle formation occurs only above:
 - **Critical Micelle Concentration (CMC)**
 - **Kraft temperature (Tk)**
- **Examples:** Soaps and synthetic detergents.

Tyndall Effect

- If a homogeneous solution is placed in the dark and observed in the direction of light, it appears clear. When observed at right angles to the light beam, it remains dark.
- Colloidal solutions may appear reasonably clear or translucent when viewed in transmitted light, but show **mild to strong opalescence** when viewed at right angles to the light beam. The path of the beam becomes visible due to scattering of light, often appearing bluish.
- The **Tyndall effect** was first observed by Faraday and later studied in detail by Tyndall. It occurs due to the **scattering of light by colloidal particles**, which illuminates the path of the beam.
- The Tyndall effect can be observed during the projection of pictures in the **cinema hall** due to the **scattering of light by dust and smoke particles** present there.
- **Conditions for Tyndall effect**
 - The diameter of dispersed particles is not much smaller than the wavelength of the light used.
 - The refractive indices of the dispersed phase and the dispersion medium differ significantly.

Suspension

- A suspension is a heterogeneous mixture in which solute particles do not dissolve but remain suspended in the medium.
- Unlike a **colloid**, which has particles in the range of **1–1000 nm**, a **suspension** contains relatively larger particles, typically **greater than 1000 nm**.

Properties of Suspension:

- It is a **heterogeneous mixture**.
- Particles are **visible to the naked eye**.
- Particles **scatter light** passing through the suspension, making the path of light visible.
- Solute particles **settle down** if the suspension is left undisturbed, making it an **unstable mixture**.

Hard Water And Soft Water

Soft water forms lather easily with soap, while **hard water** does not.

Hard Water	Soft Water
Does not produce lather with soap readily and scum formation takes place.	Produces lather with soap readily
Hardness of water is due to the presence of bicarbonates, chlorides, and sulphates of calcium (Ca^{2+}) and magnesium (Mg^{2+}) in it.	It has some quality of sodium ions only

Formed when water flows through limestone and chalk .	Produced by passing hard water over an ion exchange resin .
Has a high mineral content .	Has a low mineral content.
E.g.: Mineral Water, well water, underground & river water, high pH water, etc.	E.g.: Rain or snow water, distilled water, RO water, etc.

Types of Hardness of Water

1. Temporary Hardness (Carbonate Hardness):

- Caused by **bicarbonates of calcium and magnesium** $\rightarrow \text{Ca}(\text{HCO}_3)_2, \text{Mg}(\text{HCO}_3)_2$
- Also called **carbonate hardness**.
- **Easily removed by boiling and filtration**.

2. Permanent Hardness (Non-carbonate Hardness):

- Caused by **chlorides and sulphates of calcium and magnesium** $\rightarrow \text{CaCl}_2, \text{CaSO}_4, \text{MgCl}_2, \text{MgSO}_4$
- Also called **non-carbonate hardness**
- **Cannot be removed by boiling**
- **Removal methods:** adding slaked lime (Clark's process), adding Washing soda, process using Ion exchange resins, and calgon process.

Soap

- Soaps are sodium or potassium salts of long-chain fatty acids that act as cleansing agents in water.
- They are made from **vegetable oils or esters**. Soaps include **sodium oleate, sodium stearate and sodium palmitate**, represented as $\text{RCOO}^- \text{Na}^+$ (e.g., sodium stearate $\text{CH}_3(\text{CH}_2)_{16}\text{COO}^- \text{Na}^+$).
- Soaps are formed by heating **vegetable oil or fat (glyceryl ester of fatty acid)** with aqueous **NaOH solution**.
Vegetable oils/Esters + Alkali (NaOH) $\rightarrow$ Salt (Soap) + Glycerol

This process is called **saponification**, where esters are hydrolysed to form soap in **colloidal form**. Soap is separated using **NaCl**, and **glycerol** is recovered from the remaining solution by **fractional distillation**.

- Generally, **potassium soaps are softer to the skin than sodium soaps**. They are prepared by using **potassium hydroxide (KOH) solution** in place of **sodium hydroxide (NaOH)**.

Detergents

- Detergents are **cleaning agents** that work effectively in **both soft and hard water**, as they **form foam even in hard water and even in cold (ice) water**.
- Technically, detergents are **sodium or potassium salts of long alkyl chain ending with a sulfonate group** and are soluble even in hard water.

Types of Synthetic Detergents

Synthetic detergents are mainly classified into **three categories**:

A. Anionic Detergents

- Most widely used detergents.
- In water, form **anions (negatively charged ions)** $\rightarrow$ active cleaning part.
- Made as **sodium salts of sulphonated long-chain alcohols or hydrocarbons**.
- General structure: $\text{R-SO}_3^- \text{Na}^+ / \text{R-O-SO}_3^- \text{Na}^+$

Preparation:

- Long-chain alcohols react with concentrated sulphuric acid to form alkyl hydrogensulphates, which on neutralisation with alkali produce **anionic detergents**. Similarly, alkyl benzene sulphonic acids, when neutralised with alkali, form **alkyl benzene sulphonate detergents**.

Important type:

- Alkylbenzene sulphonates** → widely used in household detergents.

B. Cationic Detergents

- Cationic detergents are **quaternary ammonium salts of amines**, having acetates, chlorides, or bromides as anions.
- Their cationic part contains a **long hydrocarbon chain** and a positively charged nitrogen atom, hence they are called cationic detergents.
- They possess **germicidal properties**.
- Are **expensive and therefore have limited use**.

Important Type:

- Cetyltrimethylammonium bromide** is a commonly used cationic detergent → widely used in hair conditioners.

C. Non-ionic Detergents

- Do not form ions in solution.**
- Example: formed by reaction of **stearic acid with polyethylene glycol**.
- Common Type: **liquid dishwashing detergents**.

Cleaning Action:

- The cleansing action is similar to that of soaps; they remove grease and oil through micelle formation.

Chromatography

- Chromatography is a widely used technique for separating mixtures into their individual components, purifying compounds, and assessing their purity.
- In this technique, a mixture of substances is applied onto a **stationary phase**, which may be a solid or liquid. A **pure solvent or a mixture of solvents** or a gas is allowed to move slowly over the stationary phase. The components of the mixture separate gradually from one another due to differences in their movement. The moving phase is called the **mobile phase**.
- Developed by **Mikhail Tsvet (1903)**.
- Applications:**
 - Separation of drugs from blood
 - Separation of different pigments from extract of flower petals.

BCW Facts:

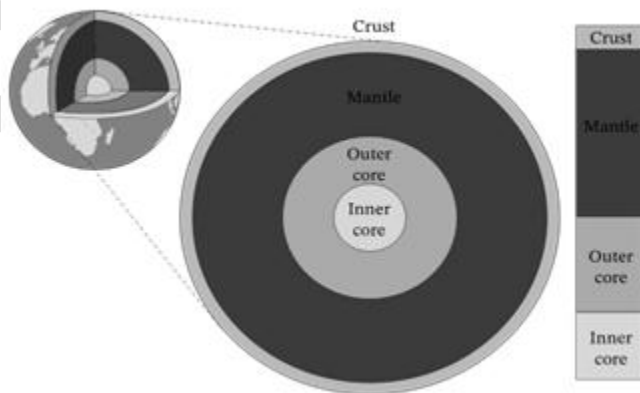
- Two solutions with the **same osmotic pressure at a specific temperature** are known as **isotonic solutions**.
- Mixtures consist of **more than one type of pure substance**. For instance, **milk is a mixture** containing water, fats, proteins, and other components.
- Soap removes grease** through a process called **emulsification**. It breaks down the grease into smaller droplets, **allowing them to mix with water and be rinsed away**.
- Soaps and detergents reduce the surface tension** of a solution, making them effective cleaning agents. Substances that lower surface tension are known as **surface-active agents or surfactants**.
- Pasteurized milk** is heated to **around 70°C for 15 to 30 seconds**, then rapidly cooled and stored. Pasteurization is the **process of heating milk to a specific temperature for a set period** to eliminate pathogenic microorganisms that may be present.

3- Minerals And Ores

Minerals are **naturally occurring inorganic substances** with a definite chemical composition and a crystalline structure. They are the building blocks of rocks and are found in diverse forms in the Earth's crust. Minerals may consist of single elements, such as gold or copper, or compounds like quartz (SiO_2) and feldspar. Ores are **specific types of minerals** that contain a high concentration of metals or valuable elements, making them economically feasible to extract.

Interior of Earth

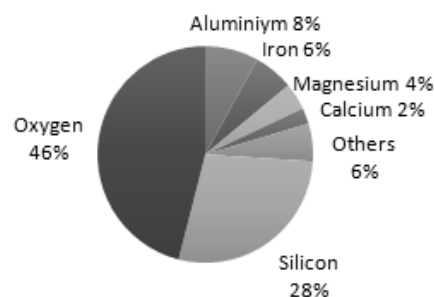
- The interior of the Earth has four main layers: the **crust, mantle, outer core, and inner core**, each differing in composition and properties. The Earth is made mainly of rocks and minerals. The **crust** is the thin, solid outer layer, the **mantle** is semi-solid, and the **core** is divided into a **liquid outer core** and a **solid inner core** due to extreme pressure.
- The Earth can be compared to an egg: the **crust**, a **silicate solid**, is like the **eggshell**, the **mantle** is like the **egg white layer**, and the **core** is like the **yolk**, with the outer core being viscous molten and the inner core dense solid.



Crust

- The outermost solid part of the earth; normally **about 8-40 km thick**.
- The **crust** is **brittle in nature**, its thickness varies across regions. The **oceanic crust** is **thinner (about 5 km)**, while the **continental crust** is **thicker (about 30 km)**.
- It is mainly composed of **silica (Si)** and **aluminium (Al)**, and is therefore also called **SIAL**.

Elemental Composition Of The Earth's Crust



Mantle

- The portion of the Earth's interior beyond the crust is called the mantle.
- The mantle is about **2900 km in thickness**. Extends from Moho's discontinuity to a depth of 2,900 km.
- It makes up nearly **84% of Earth's volume and 67% of its mass**.
- It is mainly composed of **silicon and magnesium**, hence called **SIMA**.
- The mantle is also the **main source of magma** that erupts during volcanic activity.

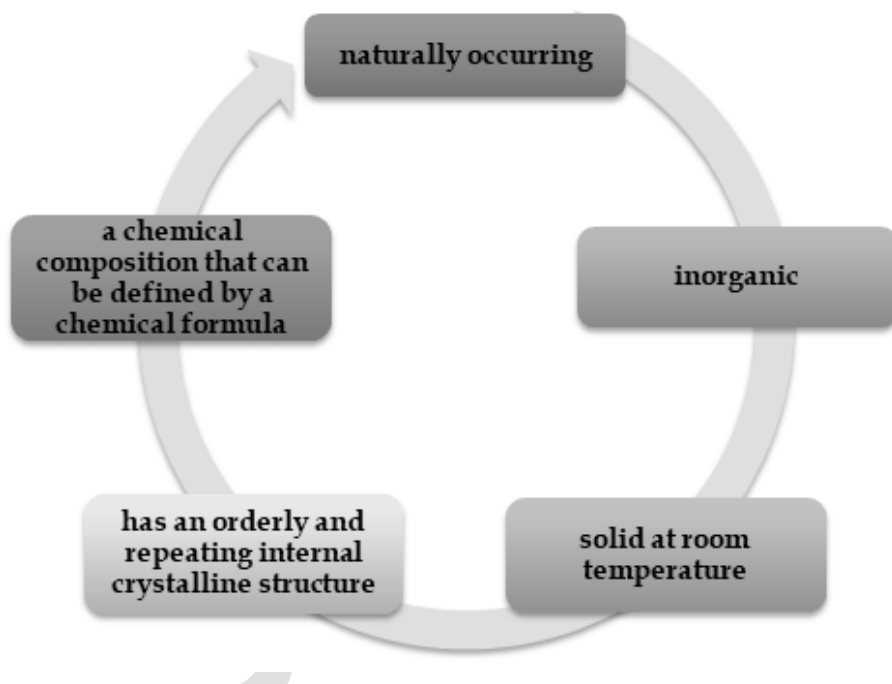
Core

- The **innermost layer** of the Earth surrounds its centre.
- Mainly composed of **nickel (Ni) and iron (Fe)**, it is therefore called **NIFE**.
- The core accounts for about **15% of Earth's volume and 32.5% of its mass**.
- It is divided into two parts: the **outer core (liquid state)** and the **inner core (solid state)**.

BCW Fact: The crust and the uppermost part of the mantle are collectively known as the **lithosphere**.

Minerals

Minerals are **inorganic substances with definite chemical and physical properties**. In geology, the classic definition of a mineral is represented below through a diagram:



This means that man-made substances, such as steel, are not minerals. **Minerals are of two types:**

- (i) Metallic (e.g., Iron, Manganese, etc)
- (ii) Non-metallic (e.g., Mica, Limestone, etc)

Metal, Non-Metal, And Metalloid

Properties	Metals	Non-Metals	Metalloids
Melting and Boiling Point	Generally high	Generally low	Intermediate
Conductivity	High	Low	Intermediate

Hardness	Hard and durable	Soft and brittle	Soft and brittle
Lustre	Shiny & reflective	Dull & non-reflective	Dull & non-reflective
Density	Dense solid structure	Less dense as compared to metals	Generally less dense than metals but denser than non-metals
Physical Forms	Solid at room temperature (except Mercury)	Solid, liquid or gas at room temperature	Solid at room temperature but easily converted to liquid or gas when heated
Ionisation	Tends to lose electrons and form positive ions	Tends to gain electrons and form negative ions	Intermediate ionization energies and electronegativity values

Metalloids

- **Metalloids** are elements found between **Group 13 and Group 17** of the periodic table.
- They possess properties (chemical properties and physical properties) intermediate between **metals and non-metals**.
- The seven metalloids are **Boron, Silicon, Germanium, Arsenic, Antimony, Tellurium, and Polonium**.
- **Boron (B)** is a metalloid with atomic number **5**. It is a **hard, low-density, refractory solid** with **low electrical conductivity** and a **high melting point**. Pure crystalline boron is a **black, lustrous semiconductor**, conducting electricity at high temperatures but acting as an insulator at low temperatures.

Important Metallurgy Process

Ores

- **Ores** are minerals from which metals can be extracted **conveniently and economically**. They usually contain a **high percentage of the desired metal** and fewer impurities.
- Ores also contain unwanted impurities called **gangue** mixed with the useful mineral.
- All ores are minerals, but not all minerals are ores.

Concentration of Ores

Concentration of ores is the process of removing **gangue (impurities)** from ores to obtain the metal-rich part called the **concentrate**. The method used depends on the **physical or chemical differences** between the ore and impurities.

Some of the important **methods of Concentration** are described below-

Gravity Separation (Hydraulic Washing)

- Based on the **difference in the specific gravities** of the **ore and gangue**. The lighter impurities are washed away, leaving behind the heavier ore particles.
- **Used for the concentration of heavier oxide ores** such as hematite (Fe_2O_3), tinstone or Cassiterite (SnO_2), and gold (Au).

Magnetic Separation

- Based on the **difference in magnetic properties** of the ore and gangue.

- **Example:** Concentration of Iron ores like magnetite (Fe_3O_4) and chromite (FeCr_2O_4).

Froth flotation

- Used for **removing gangue from sulphide ores**, such as galena (PbS) or copper pyrites CuFeS_2 .
- This method made the extraction of copper from **low-grade ores economically profitable**.

Leaching

- This process involves dissolving the ore in a suitable solvent.
- **Example:** Extraction of aluminum from bauxite (using sodium hydroxide in the Bayer process).

Extraction of Crude Metal

The extraction of crude metals from concentrated ores involves two primary processes: a) Conversion of Ores into metal oxide, and b) Reduction of the oxide to metal.

a) Conversion of Ores into metal oxide

The concentrated ore is converted into metal oxide by calcination or roasting-

Calcination

- **Calcination** involves heating the ore in the **absence or limited supply of air** to remove moisture, water of hydration, and volatile impurities and convert the ore into a metal oxide. For example, **calcium carbonate (CaCO_3)** on heating forms **calcium oxide (CaO)** and **carbon dioxide (CO_2)**.

Roasting

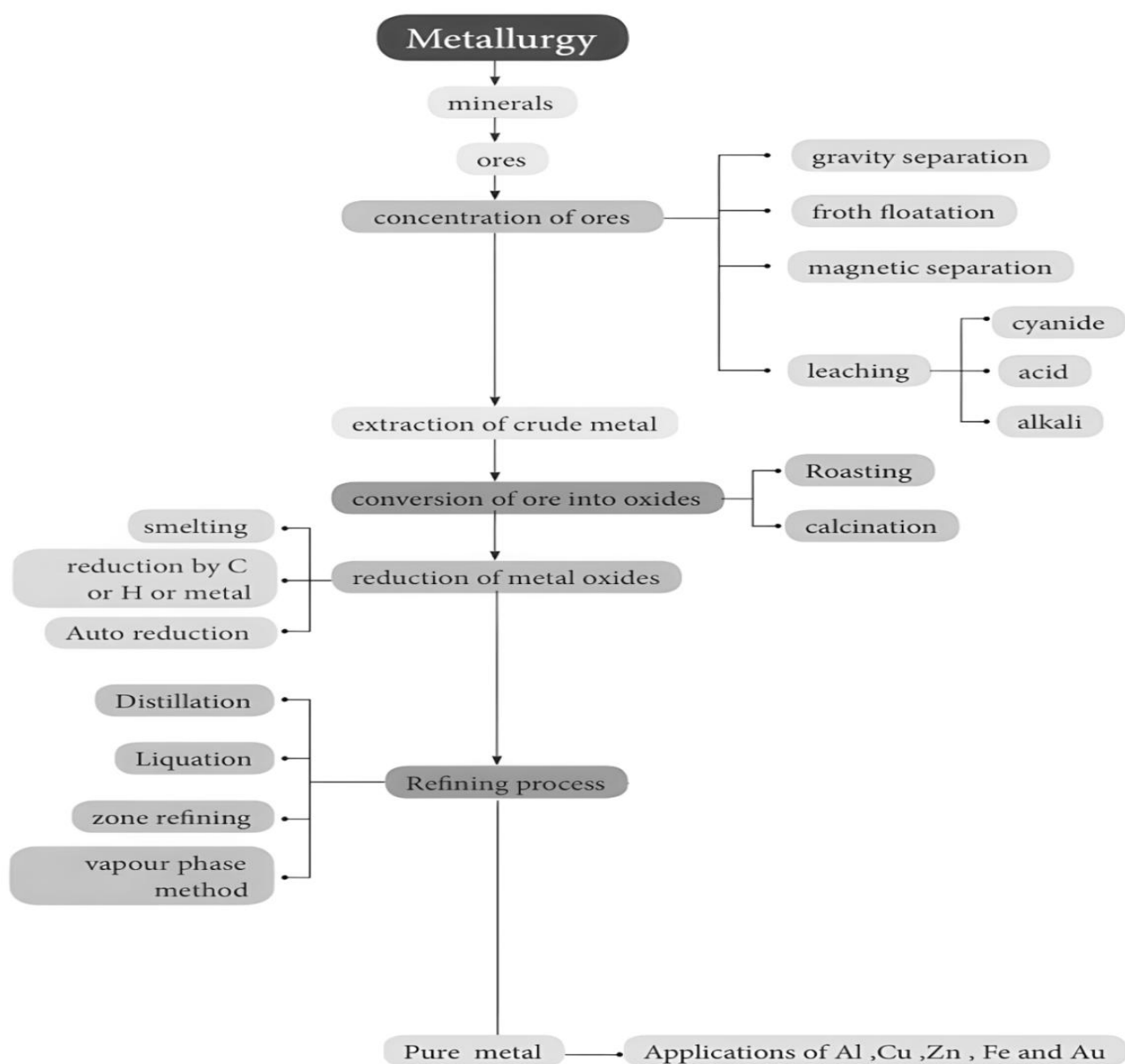
- **Roasting** involves heating the ore in the **presence of excess air** below the metal's melting point.
- Mainly used in metallurgy to **extract metal oxides** from **sulphide ores**. For example, in this process, sulphide ores of **Zn, Pb, and Cu** are converted to their respective oxides by heating them strongly in the presence of excess air.

b) Reduction of the oxide to metal

Smelting

- **Smelting** is the process of heating a **metal oxide with a reducing agent and flux** in a furnace to obtain the metal in **molten (fused) form**.
- It is a method of **extractive metallurgy** used to obtain metals such as **iron, copper, and silver** from their ores.
- During smelting, impurities are removed as **gases or slag**, leaving behind the metal.

Process Of Metal Extraction



Iron

Iron ores are rocks and minerals from which **metallic iron can be extracted**. India has large iron ore reserves. Some of the common ores of Iron are:

A. Magnetite

- Most important and best type of iron ore. It contains about **72% metallic iron and is reddish and blackish in colour**, therefore called as the **black ore of Iron**.

B. Hematite

- Contains about **60-70% metallic iron** and is **red and brown** in colour.
- The **most important industrial iron ore**, due to its extensive use, although its iron content is slightly lower than that of magnetite.

C. Limonite

- Contains about **30-40% metallic iron**. It is mostly brown in colour and is considered a **low-grade iron ore**.

D. Siderite

- It is complete inferior quality with less than **40% iron**. It has **maximum impurities and mining** is not economically viable.

Iron Ore and Commercial Forms of Iron

- Iron pyrite (FeS_2)**, also called **fool's gold** (whose appearance is like gold) and is a naturally occurring **iron disulphide mineral**. It is the most abundant sulphide mineral.
- Wrought iron or malleable iron** is the purest commercial form of iron, containing less than **0.08% carbon**. It is produced from cast iron by oxidising impurities in a **reverberatory furnace lined with haematite**. It is **soft, ductile, magnetic**, and has high **tensile strength**, making it suitable for **wires, bolts, chains, and agricultural implements**.
- Cast iron** is made by melting pig iron with scrap iron and coke using a hot air blast. It is used for casting stoves, railway sleepers, toys, etc.
- Pig iron** is a **hard and brittle form of iron** with high carbon content, obtained in a **blast furnace** by reducing iron oxide with carbon. Some of the carbon gets mixed with pure iron, making it brittle and hard.

Aluminium

Ore	Chemical Name	Formula
1. Bauxite	Hydrated aluminium oxide	$\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$
2. Cryolite	Sodium aluminium fluoride	Na_3AlF_6
3. Corundum	Anhydrous aluminium oxide	Al_2O_3

A. Bauxite

- Bauxite is the principal ore of aluminium** and contains approximately 60% aluminium oxide.
- Bauxite forms through the decomposition of rocks, resulting in a residual mass of weathered material.
- In 1821, the **French geologist Pierre Berthier** discovered bauxite.

B. Cryolite

- Cryolite is a salt of **sodium aluminium hexafluoride, represented as Na_3AlF_6** . It consists of **aluminium, sodium and fluorine**. Cryolite ores are the chief ore of aluminium.

C. Corundum

- Corundum** is a crystalline form of **aluminium oxide (Al_2O_3)**.
- Naturally transparent but may show different colours due to the presence of **transition metal impurities**.
- The two main gem varieties of corundum are **ruby** and **sapphire**.

- **Rubies** → red colour (due to the presence of **chromium**)
- **Sapphires** → various colours depending on the metal impurities present. A rare variety, the **padparadscha sapphire** → pink-orange in colour.

Extraction of Aluminium

Aluminium is obtained by **electrolysis of alumina**. The concentration of the bauxite ore into alumina is done by **Bayer's process**. Alumina is then refined into pure aluminium metal using the **Hall electrolytic process**.

Metallurgy

1. Aluminium

- **Ores:** Bauxite, Cryolite, Corundum.
- **Extraction Process:** Bayer's Process and Hall's Process
- **Physical Properties:** Silvery-white metal, malleable and ductile, low density, and a good conductor of heat and electricity
- **Chemical Properties:** Reacts with air, water, alkalis; acts as a reducing agent.
- **Uses:** Home appliances, electrical appliances, aeronautics

2. Copper

- **Ores:** Copper pyrites, Copper glance, Cuprite (ruby copper)
- **Extraction Process:** Concentration of ore, roasting, smelting, Bessemerisation, refining
- **Physical Properties:** Reddish-brown metal, high luster, dense, and malleable
- **Chemical Properties:** Reacts with oxygen on heating, chlorine, and certain acids.
- **Uses:** Electric cables and electrical appliances, utensils and containers, and coins and electroplating

3. Iron

- **Ores:** Haematite, Magnetite, Iron pyrite
- **Extraction Process:** Concentration (gravity separation), roasting, calcination, and smelting
- **Physical Properties:** Greyish-white metal, lustrous, high tensile strength, ductile
- **Chemical Properties:** Reacts with air, moisture, steam, chlorine, and acids
- **Uses:** radiators, railings, drain pipes; springs, anchors, chains.

Ores Of The Important Elements

Metal	Ore	Composition
Aluminium	Bauxite	$\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O}$
	Diaspore	$\text{Al}_2\text{O}_3 \cdot \text{H}_2\text{O}$
	Kaolinite	$\text{Al}_2\text{Si}_2\text{O}_5(\text{OH})_4$
Iron	Haematite	Fe_2O_3
	Magnetite	Fe_3O_4
	Siderite	FeCO_3
	Limonite	$\text{Fe}_2\text{O}_3 \cdot 3\text{H}_2\text{O}$
Copper	Copper pyrite	CuFeS_2
	Copper glance	Cu_2S
	Cuprite	Cu_2O
	Malachite	$\text{Cu}_2\text{CO}_3(\text{OH})_2$
	Azurite	$2\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$
Zinc	Zinc blende or Sphalerite	ZnS
	Calamine	ZnCO_3
	Zincite	ZnO

Lead	Galena	PbS
	Anglesite	PbSO ₄
	Cerussite	PbCO ₃
Tin	Cassiterite (Tin stone)	SnO ₂
Silver	Silver glance (Argentite)	Ag ₂ S
	Pyrargyrite (Ruby silver)	Ag ₃ SbS ₃
	Chlorargyrite (Horn Silver)	AgCl
	Stephanite	Ag ₂ SbS ₄
Mercury	Cinnabar	HgS

BCW Facts:

- **Graphite and iodine** possess metallic lustre though they are non-metals.
- The **Minas Gerais province of Brazil** is one of the leading iron ore producers.
- **Ilmenite (FeO.TiO₂)** and **rutile (TiO₂)** are the primary minerals of **titanium**.
- Most toothpastes are whitened using **titanium dioxide**, derived from minerals such as **rutile, ilmenite, and anatase**. Additionally, the sparkle in certain toothpastes is created using **mica**.

4- ACIDS, BASES, AND SALTS

Acids, bases, and salts are important chemical substances widely used in nature, daily life, and industries. **Acids** generally have a **sour taste** and release **hydrogen ions (H⁺)**, while **bases** are **bitter and slippery** and release **hydroxide ions (OH⁻)**. **Salts** are formed by the reaction between acids and bases and have many everyday applications.

The **pH scale** measures the acidity or basicity of a solution and helps determine its strength. Acids, bases, and salts are found in food, cleaning products, industries, and biological processes, making them essential to life and chemistry.

ACIDS AND BASES

Acids

- The word **acid** is derived from the Latin word **acere**, meaning **sour**. Natural acids are found in substances such as **lemon juice, orange juice, curd, and vinegar**.
- In water solution, it **tastes sour, turns blue litmus red, and reacts with bases to form salts**. Acids also react with some metals (e.g., iron) to release **hydrogen gas**.
- Chemically, an acid is a **hydrogen-containing substance capable of donating a proton (H⁺ ion)** to another substance.

Bases

- In chemistry, a **base** is a chemical species that **donates electrons, accepts protons, or releases hydroxide (OH⁻) ions** into an aqueous solution.



Carbonic acid
Aerated drinks



Sulphuric acid
Car batteries & paint



Tartaric acid
Wine manufacturing



Nitric acid
Fertilizers



Hydrochloric acid
Digestion



Salicylic acid
Aspirin



Acetic acid
Vinegar

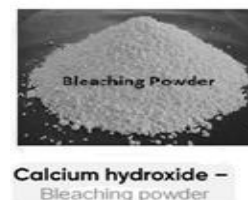


Citric acid
Citrus fruits



Acids used in
Labs

- Bases exhibit some specific properties that can be used to help identify them. Some of these properties include **slippery (soapy) to the touch, taste bitter, corrosive in nature, the ability to change the colour of the indicators (e.g., turning red litmus paper blue)**, the tendency to react with acids to form salts, and their role in promoting certain chemical reactions (**base catalysis**).

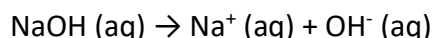


General Concepts Of Acids And Bases

Arrhenius Concept:

- According to this theory, **acids** are substances that **release hydrogen ions (H^+)** in aqueous solutions, while **bases** are substances that **release hydroxide ions (OH^-)** in aqueous solutions.
- Example:** Hydrochloric acid (HCl), an example of Arrhenius acid.
 $HCl(aq) \rightarrow H^+(aq) + Cl^-(aq)$

Sodium hydroxide (NaOH), an example of Arrhenius base.



Brönsted-Lowry Concept:

- According to the Brönsted-Lowry theory, an **acid** is a substance that can **donate a hydrogen ion (H^+)**, while a **base** is a substance that can **accept a hydrogen ion (H^+)**. In summary, acids are **proton donors** and bases are **proton acceptors**.
- Example:** In the reaction between ammonia (NH_3) and water, ammonia accepts a proton from water, making it a base and leaving water as the acid.

Lewis Concept:

- In this concept, **acids are electron pair acceptors**, and **bases are electron pair donors**.
- For example**, a molecule like boron trifluoride (BF_3) is a Lewis acid because it can accept an electron pair, while ammonia (NH_3) is a Lewis base because it can donate an electron pair.

Chemical Properties Of Acids And Bases

A. Reaction with indicators:

- Acid-base indicators** such as **litmus** and **phenolphthalein** help identify whether a substance is **acidic or basic** by changing colour.
- Acid-Base indicators (also known as pH indicators)** are substances that change colour with pH. They are usually **weak acids or bases** that, when dissolved in water, **dissociate slightly and form ions**.

Indicator	Original Colour (At Ph 7)	Colour In Acid (Ph < 7)	Colour In Base (Ph > 7)
Red litmus	Red	Red (No change)	Blue
Blue litmus	Blue	Red	Blue (No change)
Red cabbage water	Purple	Red, pink	Blue, green, yellow

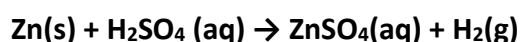
Turmeric water	Yellow	Yellow (No change)	Red
Phenolphthalein	Colourless	Colourless (No change)	Pink, red

BCW Facts:

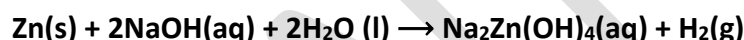
- **Phenolphthalein and methyl orange** are synthetic indicators, while litmus, turmeric water, and red cabbage are natural indicators.
- **Litmus** solution is extracted from **lichens**.
- **Olfactory Indicators:** Substances whose odour changes in acidic or basic media are called olfactory indicators. E.g., cloves, onions, vanilla extract.

B. Reaction with metals:

- When an acid reacts with a metal, hydrogen gas is evolved, and a corresponding salt is formed.
- For example, zinc metal reacts with sulphuric acid, producing zinc sulphate and hydrogen gas.



- When a base reacts with a metal, hydrogen gas is evolved along with the formation of salt that contains a negative ion composed of the metal and oxygen.
- For example, Zinc reacts with Sodium hydroxide (NaOH) in the presence of water (H₂O) to form sodium zincate (Na₂Zn(OH)₄), along with the evolution of hydrogen gas.

**C. Reaction with metal carbonates and metal hydrogen carbonates:**

- All metal carbonates or hydrogen carbonates react with acids to give a salt, carbon dioxide and water.



The **CO₂ evolved** in the above reactions, when passed through lime water (**Ca(OH)₂ solution**), turns the **lime water milky** due to the formation of white coloured precipitate of **calcium carbonate (CaCO₃)**.



NOTE: Calcium oxide reacts vigorously with water to produce slaked lime, releasing large amounts of heat. It is also a type of combination reaction.

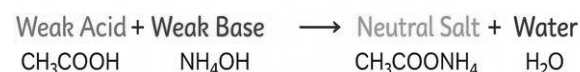
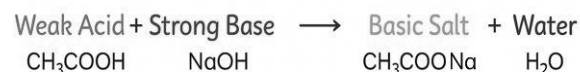
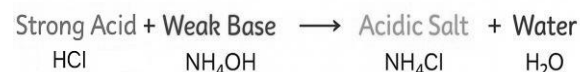
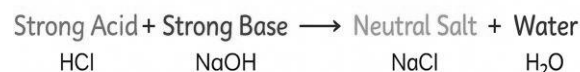
**D. Acid - Base Reaction (Neutralisation Reaction):**

- The reaction between **an acid and a base to give a salt and water** is known as **a neutralisation reaction**.



- The **pH of a neutralized solution** depends on the **strengths of the acid and base** involved.

- A **strong acid + strong base** forms a **neutral salt**.
- A **strong acid + weak base** forms an **acidic salt** due to hydrolysis of the cation from the weak base.
- A **weak acid + strong base** forms a **basic salt** due to hydrolysis of the anion from the weak acid.



Application of Neutralization Reaction in Our Day-to-Day Life

Activity	Solution
Treatment of bee stings	Bee stings contain formic acid and are acidic in nature, these stings can be neutralized by applying a base like moist baking soda (NaHCO_3) or calamine solution (contains ZnCO_3) .
Treatment of tooth decay or cavities	Acids produced in the mouth by bacteria acting on food particles reacts with enamel i.e., calcium hydroxyapatite and leads to cavities. Toothpaste is mildly basic and helps neutralize these acids.
To treat acidic or basic soils	Plants do not grow well if the soil is too acidic or too alkaline. Alkalis like burnt wood ash, CaO, CaCO_3, etc. are added to neutralize the acidic soil.
Treatment of acidity or gastric patients	Our stomach produces hydrochloric acid which aids in digestion of food. Anti-acids or antacids are medicines containing bases such as NaHCO_3 (sodium bicarbonate), $\text{Mg}(\text{OH})_2$ (Milk of Magnesia) , which are used to neutralize the excess acid in the stomach.

E. Reaction of metallic oxides with acids:

Metallic oxides, which are **basic** in nature, **react with acids to give salts and water**.

**F. Reaction of a non-metallic oxide with base:**

Non-metallic oxides, which are usually **acidic** in nature, react with an alkali, leading to the formation of the corresponding salt and water.



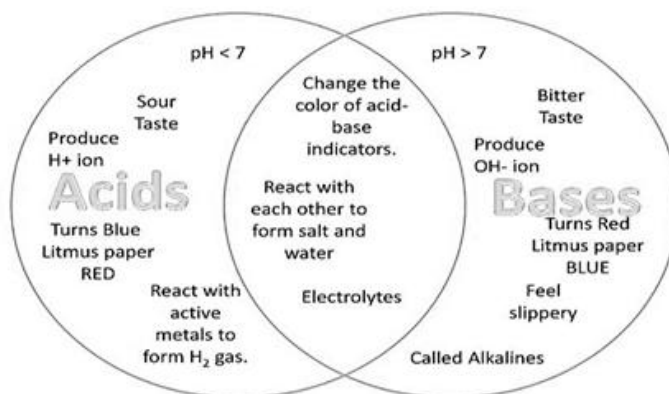
Characteristics Of Acids And Bases

- Acids **dissociate into positive hydrogen ions**, while **bases can dissociate into negative hydroxide ions**.



Acids release H^+ ions, which combine with water molecule to form hydronium, H_3O^+ ions. The extent of their dissociation determines whether an acid or base is **strong** or **weak**.

- Strong Acids:** Acids that give rise to more H^+ ions. E.g., HCl , HNO_3
- Weak Acids:** Acids that give fewer H^+ ions. E.g., CH_3COOH
- When a base is mixed with water, the following reaction occurs:
Bases have the **capacity to liberate hydroxide (OH^-) ions**, so they **dissociate into hydroxide anions** and the cation of the salt.



- For example: $\text{NaOH} + \text{H}_2\text{O} \rightarrow \text{Na}^+ + \text{OH}^-$
- An example of a **strong base** is NaOH , and a **weak Base** is NH_4OH .

Hence, acids and bases mix in water to dissociate and form hydronium and hydroxide ions, respectively.

- **Bases that are soluble in water are called Alkalis.**

Naturally Occuring Acids	
Name of the Acid	Natural Source
Lactic acid	Curd
Acetic acid	Vinegar
Citric acid	Citrus fruits such as oranges, lemons, etc.
Tartaric acid	Tamarind, Grapes
Oxalic acid	Tomato and Spinach
Methanoic acid (or Formic Acid)	Ant's sting, Bee's sting, and Nettle sting

BCW Fact: **Nettle**, a herbaceous plant, has fine stinging hairs that inject **Methanoic acid** into the body of any animal or human that comes into contact with it.

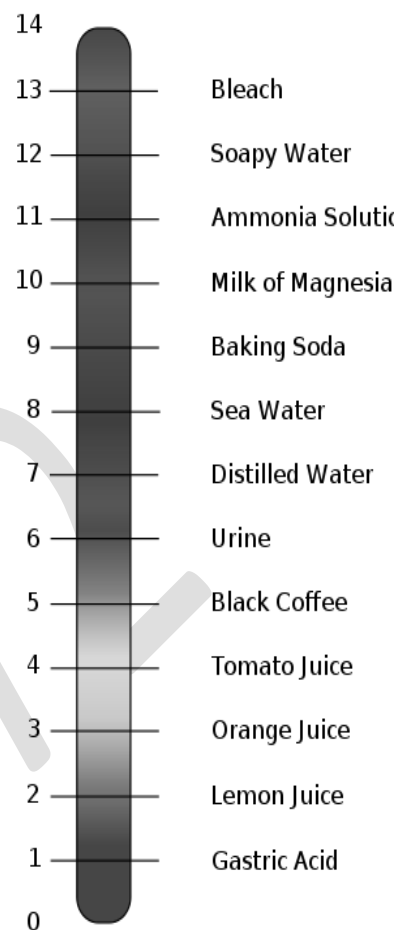
Potential Of Hydrogen (pH)

- The pH (potential of hydrogen) scale is used to **specify the acidity or basicity of an aqueous solution**. It was devised by the Danish biochemist Soren Sorensen in 1909.
- pH is defined as the negative log of the **concentration of H^+ ions**.
- Acidic solutions have lower pH values than basic or alkaline solutions.
- The “p” in pH stands for '**potenz**' in German, meaning power. The range goes from **0 - 14, with 7 being neutral**.
- **pH values less than 7** indicate **acidity**, whereas a **pH of greater than 7** indicates a **base**.
- pH is a measure of the relative amount of free hydrogen and hydroxyl ions in the water.
- The **Human body** works within the pH range of **7.0 to 7.8**.

- When the pH of **rainwater** falls **below 5.6**, it is called **acid rain**.

The pH of common substances:

- Gastric Juices : About 1.2 (strong acid)
- Lemon Juice : About 2.2
- Saliva before meal : About 7.4
- Saliva after a meal : About 5.8 (mild acid)
- Carrot juice : About 6
- Pure water : 7
- Neutral Solution : 7
- Human Blood (Plasma) : 7.4
- Milk of magnesia : About 10 (mild base)
- NaOH solution : About 14 (strong base)
- Lime Water : 10.5
- Milk : 6.8
- Soft drinks and vinegar : About 3.0



Acid Rain

- Acid rain is caused by a chemical reaction **when compounds like sulphur dioxide (SO₂) and nitrogen oxides (NO_x) are released into the atmosphere**. These gases rise very high in the atmosphere and react with oxygen and water to form acidic pollutants, resulting in acid rain. These gases are easily soluble in water and form **sulphuric acid** and **Nitric acid**.
- Acid rain** is harmful to **crops, rivers, trees, monuments, and buildings**.
- It corrodes the marble of the Taj Mahal, a phenomenon known as **"Marble Cancer."**
- The term acid rain was coined by **Robert Angus Smith** in 1852.

Salts

- Salts are **ionic compounds** produced **by the neutralization reaction between an acid and a base**. They are electrically neutral. Among the various salts, sodium chloride is the most common.
- Sodium chloride, also known as table salt or common salt, is widely used to enhance the taste of food.

Characteristics of salt

- Most salts are **crystalline solids** and are soluble in water.
- Salts may be **transparent or opaque**.
- Solutions of the salt **conduct electricity** even in their molten state.
- Can have various tastes, such as salty, sour, sweet, bitter, and umami (savory). Neutral salts are odourless.
- Can be colourless or coloured.

Some Important Chemical Compounds

Sulphuric Acid (H₂SO₄)

- **Sulphuric acid**, used in **car batteries**, consists of 30% - 50% of sulphuric acid in water.
- A mixture of sulphuric acid and water is sometimes used as an electrolyte in lead-acid batteries.
- Other names of sulphuric acid are - oil of vitriol and spirit of vitriol.
- Sulphuric acid is a **component of acid rain**.
- A solution of Sodium bicarbonate and dilute sulphuric acid is used in **Soda-acid fire extinguishers**.

Sodium Chloride (NaCl)

- Commonly called **common salt**, is mainly obtained by the **evaporation of seawater**.
- It also occurs as **rock salt** in underground deposits.
- Common salt is purified by **crystallisation**.
- It is an essential component of food and is used in the manufacture of **sodium hydroxide, baking soda, and washing soda**.

Sodium hydroxide (NaOH)

- Also known as **caustic soda**.
- In its pure form, it is a **crystalline solid** and is colourless, highly corrosive, and water-soluble ionic compound.
- It is prepared by the Chlor-alkali process.
 - **Chlor-alkali Process:** Electricity is passed through an aqueous solution of Sodium chloride (called brine). Sodium chloride decomposes to form sodium hydroxide. Chlorine gas is formed at the anode, and hydrogen gas at the cathode. Sodium hydroxide solution is formed near the cathode.

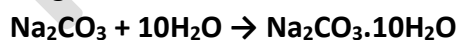


Baking soda (NaHCO₃)

- **Chemical Name:** Sodium hydrogen carbonate / Sodium bicarbonate
- Also known as cooking soda and is used for making Baking powder.
- Used as an antacid, and also used in Soda-acid fire extinguishers.

Washing soda (Na₂CO₃·10H₂O)

- **Chemical Name:** Sodium Carbonate decahydrate
- Sodium carbonate is also known as Soda ash.
- Adding water to sodium carbonate and allowing this mixture to cool produces decahydrated sodium carbonate, commonly known as washing soda.



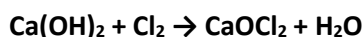
- In general, sodium carbonate is prepared by passing CO₂ gas through concentrated NaOH.



- **Uses of Sodium Carbonate:** Sodium carbonate is used in the manufacturing of glass, cleansing agents, soap, and paper. It is also used to **soften hard water**.

Bleaching powder (CaOCl₂)

- **Chemical Name:** Calcium oxychloride.
- Production takes place at the Hasenclever plant utilizing the **Bachmann process**.
- It is prepared by the reaction between chlorine and dry slaked lime at about 40⁰ C.



- It is a **strong oxidising agent** and is widely used for **bleaching**.

Uses of Bleaching powder:

- Used for bleaching **cotton, linen, and wood pulp**.
- Coloured matter + Bleaching powder → Colourless product

- Used for **disinfecting drinking water**.

Plaster of Paris ($\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$):

- Plaster of Paris (PoP) is chemically called **calcium sulphate hemihydrate**. It is named PoP because earlier it was extensively mined from Montmartre in Paris.
- It is prepared by heating gypsum at 373K.



- Used in **bandage, proofing material, and sealing agent**.
- Used for **making statues, toys, and decorative articles**.
- Used for **smoothing wall surfaces**.

BCW Facts

- Aqua regia** is a freshly prepared **mixture of concentrated hydrochloric acid and concentrated nitric acid in a 3:1 ratio**. It is a highly corrosive, fuming liquid and is one of the few reagents capable of dissolving gold and platinum. Aqua regia is also known as "**royal water**".
- The **atmosphere of Venus** consists of thick white and yellowish clouds made of **sulphuric acid**.
- Epsom Salt is the common name for **magnesium sulphate (MgSO_4)**. It is frequently used to alleviate inflammation, ease skin issues, and ease muscle soreness.
- Iodised salt** is chemically identical to common salt, but it contains small amounts of **potassium iodide** to provide iodine in the diet.

5- Carbon And Organic Compounds

Carbon is a **remarkable element** forming the basis of all living organisms. Its ability to form strong bonds with a wide variety of elements, including itself, gives it exceptional versatility. Although the Earth's crust contains only **0.02% carbon** (as minerals like **carbonates, hydrogen carbonates, coal, and petroleum**) and the atmosphere has just **0.03% carbon dioxide**, its **importance is immense**. **Organic compounds**, which are carbon-based, are essential components of life. They play **vital roles in biological processes** and have wide applications in **industries and environmental systems**.

Carbon Bonding

- Carbon (C) appears in **the second row of the periodic table and has four bonding electrons in its valence shell**.
- Like other non-metals, **carbon needs eight electrons to fill its valence shell completely**.

Atom	Carbon	Nitrogen
Atomic Number	6	7
$2n^2$ ($n=1$)	2 ($1s^2$)	2 ($1s^2$)
$2n^2$ (for $n=2$)	4 ($2s^2 2p^2$)	5 ($2s^2 2p^3$)

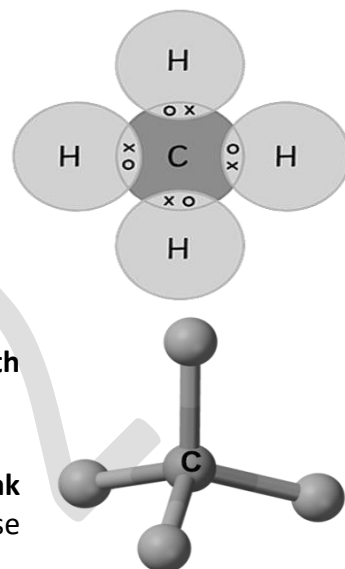
[Maximum e^- possible in valence shell =8]Valency = $8-4 = 4$ Valency = $8-5 = 3$

Nature Of Carbon Bonding

- Bonding in carbon is **covalent in nature**. Covalent bonds are formed by the sharing of electrons between two atoms so that both can achieve a filled outermost shell.
- Carbon **neither gains nor loses four electrons to complete its octet**.
- Since the carbon atom is **so small**, it would be difficult for the nucleus to hold the four extra electrons upon gaining them.
- When carbon loses its electrons, it takes a lot of energy to remove four electrons.
- By sharing its valence electrons**, carbon forms four covalent bonds and **acquires a noble gas structure**. These bonds form a **tetrahedron (a pyramid with a spike at the top)**.

Some Important Properties

- Covalently bonded molecules have strong bonds within the molecule, but **weak intermolecular forces**. This results in **low melting and boiling points** of these compounds.
- Most carbon compounds are **poor conductors of electricity**.



Allotropes of Carbon

Allotropy is the property of some elements to **exist in two or more different physical states but have similar chemical properties**.

Three main Allotropes of Carbon:

A. Diamond

- In this form of carbon, an atom is bonded to four other **carbon atoms arranged in a tetrahedral lattice (forms 3-Dimensional structures)**.
- Directional covalent bonds** are distributed consistently throughout the lattice.
- An **extremely hard, transparent, and poor electrical conductor with excellent thermal conductivity**.
- Used for drilling rocks, and cutting and making jewellery.

B. Graphite

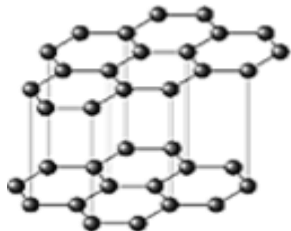
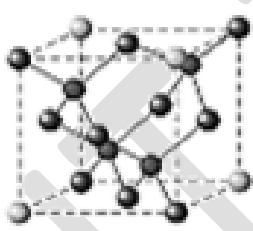
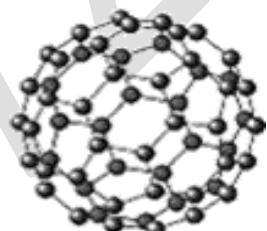
- Formed **with two-dimensional layers of carbon** atoms. Each carbon atom is linked to three carbon atoms via a C-C covalent bond, which is arranged hexagonally.
- Due to this arrangement, **layers are stacked over each other** and held by **weak Van der Waals forces**, making them **slippery and able to slide**, hence a **good lubricant**.
- It is a very **good conductor of heat and electricity**. Each carbon atom in a **hexagonal ring is sp^2 hybridised**, forming **three covalent bonds**. The **fourth electron remains free**. These **free electrons move between layers**, enabling **electrical conductivity**.

C. Buckminsterfullerene

- Fullerenes are produced by heating graphite in an electric arc **in the presence of** inert gases like helium or argon.
- An allotrope of **carbon** made of **60 carbon atoms** arranged in a **spherical, cage-like structure**, resembling a **football**. It is a **dark solid at room temperature** and is named after architect **Buckminster Fuller** due to its similarity to his **geodesic dome design**.

- **C₆₀ is made up of 20 hexagons and 12 Pentagons.** All the carbon atoms are equal, and they undergo **Sp² hybridisation**.
- It is used as a **lubricant** and **catalyst**, and has **biomedical applications** such as **MRI contrast agents and X-ray imaging agents**.
- Normally, it is a poor electrical conductor, but can exhibit conductivity when chemically modified or doped with certain elements (e.g., alkali metals like potassium or rubidium).
- **Fullerenes (C₆₀)** and their derivatives have **potential anti-viral activity**, which has strong implications for the **treatment of HIV infection**.

BCW Fact: “Fullerenes” are a pure **form of Carbon**, and it was discovered by **Robert F. Curl, Harold W. Kroto** and **Richard E. Smalley** in 1985. They were awarded the **Nobel Prize in chemistry** in 1996 for their discovery of fullerenes.

	Graphite	Diamond	Fullerene
Diagram			
Structure	Trigonal planar	Tetrahedral	Spherical
Electrical conductivity	Yes	No	No
Thermal conductivity	High	High	Low
Type of hybridization	Sp ²	Sp ³	Sp ²
Appearance	Dark grey with metallic lustre	Colourless (Coloured, if impure)	Black solid
Special properties	Insoluble in water, Soft and slippery	Insoluble in water, Brittle, Very hard substance	Do not dissolve in water, Soluble in some non-polar solvents like benzene, low melting point
Uses	Pencil lead, lubricant, making electrodes	Jewellery, drilling	Gene and drug carriers (inclusion complexes)

BCW Fact: Various other forms of elemental carbon, such as **carbon black, coke, and charcoal**, represent impure variants of graphite or fullerenes. **Carbon black** is produced through the combustion of hydrocarbons in a restricted air environment. In contrast, **charcoal and coke** are derived from the thermal decomposition of wood and coal, respectively, at elevated temperatures without the presence of air.

Versatile Nature of Carbon

Carbon forms a large variety of compounds due to its **tetravalency** and **catenation property**.

- **Tetravalent Nature:** Carbon has a **valency of 4**, enabling it to form bonds with four other atoms of carbon or other hetero-atoms through **single, double, or triple covalent bonds**.
- **Catenation:** It is the **self-linking property of carbon atoms** through covalent bonds, forming **long straight chains, branched structures, or rings of different sizes**. This is due to:
 - Small size of carbon atom
 - Strong carbon-carbon bond
 - Tetravalent nature

Prominent Features of Versatility

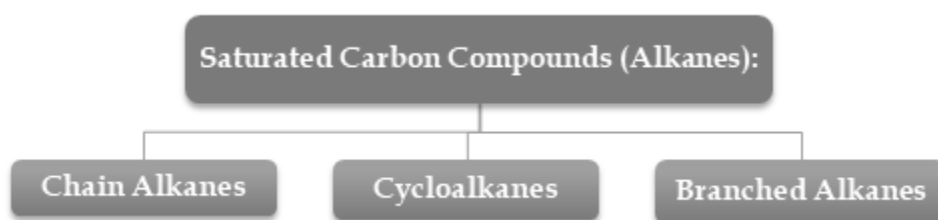
- **Single, Double & Triple Bonds.**
- **Covalent Bond Formation**
- **Homologous Series:** Formation of carbon chains leads to a **homologous series**, where compounds have the same **functional group but different chain lengths**.
- **Functional Groups:** Groups like **alcohol, aldehyde, ketone, and carboxylic acid** give **specific properties** to carbon compounds that contain them.

Saturated And Unsaturated Carbon Compounds

Saturated Carbon Compounds (Alkanes)

Alkanes are saturated open chain hydrocarbons **containing carbon - carbon single bonds**. Methane (CH_4) is the first member of this family.

The formula for Alkanes is $\text{C}_n\text{H}_{2n+2}$. They are classified into three types:



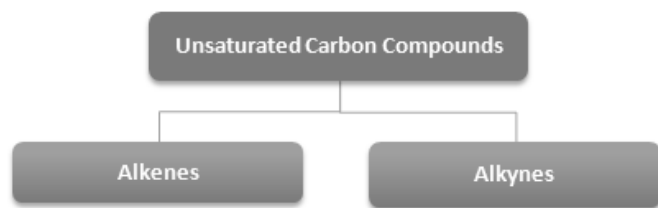
- **Alkanes are colourless and odourless.** They have **weak Van Der Waals forces** of attraction. Initial members of alkanes are gases, intermediate members are liquids and higher alkanes are solids.
- **Shorter chain alkanes have low melting and boiling points.**
- **Alkanes are non-polar molecules** because of the covalent bond, hence it is non soluble in water.
- Alkanes have a **lower density than water**, hence they float on water.

Chemical Properties:

- **Combustion of alkanes:** In excess of oxygen O_2 , alkanes readily undergo combustion producing carbon dioxide gas, water, and energy in the form of heat and light. Alkanes combustion produces clean flames.
- **Halogenation:** Alkanes are less reactive. Without ultraviolet light, they do not react with halogens. With UV light, halogenated alkane is produced.
- **Propane and butane are used in propane gas burners**, as propellants or aerosol sprays when liquefied at low temperatures.

Unsaturated Carbon Compounds

- Due to **lack of sufficient number of hydrogen to form the required number of C-H bonds**, $C=C$ or $C\equiv C$ is formed. They are more reactive than the saturated carbon compounds.



Alkenes

- The hydrocarbons that contain at **least one double bond between two adjacent carbon atoms** are called **alkenes**. The general formula of Alkenes is C_nH_{2n} and the smallest alkene is **ethylene (or ethene)**.
- The first **three members are gases**, the next fourteen are liquids and the higher ones are solids.
- Ethene is a colourless gas with a faint sweet smell. All other alkenes are colourless and odourless, insoluble in water but soluble in non-polar solvents like benzene and petroleum.
- They show a **regular increase in boiling point and melting point** with an increase in size.

Alkynes

- The hydrocarbons that contain at least two carbon atoms connected by a **triple bond**. Its **general formula is C_nH_{2n-2}** . The smallest alkyne is acetylene (ethyne) with the chemical formula C_2H_2 .
- The first three members are gases, the next eight are liquids and the higher ones are solids.
- All **alkynes are colourless**. Ethylene has a characteristic sweet and musky odour. Other members are odourless.
- Alkynes are weakly polar in nature. Their melting point, boiling point and density increase with the increase in molar mass.

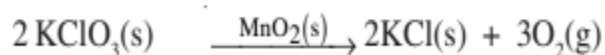
BCW Fact: Acetylene is used for arc **welding** in the form of oxyacetylene flame obtained by mixing acetylene with oxygen gas.

Ethylene is a natural, safe, and necessary **phyto-hormone for ripening**. When fruits are artificially ripened, they are assisted by the use of ethylene gas. Fruits are kept in specially designed gas chambers and exposed to ethylene.

- Combustion:** Combustion of Unsaturated hydrocarbons produces **yellow flames and black smoke**.
- Hydrogenation of Alkenes:** Unsaturated hydrocarbons such as alkenes add hydrogen in the presence of catalysts such as platinum, palladium or nickel to give saturated hydrocarbons. This reaction is commonly used in the **hydrogenation of vegetable oils, a process that involves the addition of hydrogen gas to convert them into vanaspati ghee**, using a **nickel** catalyst.

Catalysis

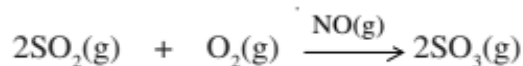
- A catalyst is defined as a substance that alters the rate of a chemical reaction while remaining chemically unchanged upon completion of the reaction.
- For example, manganese dioxide (MnO_2) significantly accelerates the following reaction.

**Types of Catalysts:**

Catalysts can be categorized into various types:

1. **Homogeneous Catalysts:** These catalysts are present in the same phase (solid, liquid, or gas) as the reactants.

Example: Nitric oxide facilitates the oxidation of sulphur dioxide to sulphur trioxide in the lead chamber process.



2. **Heterogeneous Catalysts:** These catalysts are found in a different phase from the reactants.

Example: Iron (Fe) is utilized in the Haber process for the synthesis of ammonia.



3. **Enzymes (Biocatalysts):** These are biological catalysts, primarily proteins that enhance the rate of biochemical reactions.

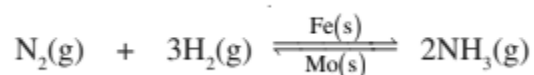
Example: Amylase assists in the conversion of starch into maltose.

Inhibitor: An inhibitor is a chemical or biological molecule that slows down or halts the rate of a chemical reaction.

Example: Glycerol retards the decomposition of hydrogen peroxide.

Catalytic Promoters:

- These are substances that increase the activity of a catalyst without displaying any catalytic activity.
- **Example:** In the Haber process for ammonia production, molybdenum, which serves as a promoter, augments iron's catalytic activity.

**Catalytic Poisons:**

- These substances decrease or completely obstruct the function of a catalyst without displaying any catalytic activity.
- **Example:** In the contact process for sulphuric acid production, even minute amounts of arsenious oxide (As_2O_3) can poison the platinum catalyst. Therefore, vanadium pentoxide (V_2O_5) is now favoured as the catalyst due to its resilience against arsenic poisoning.

Oxidation And Reduction

Oxidation and Reduction reactions can be characterized in two distinct ways: **(i) in terms of loss or gain of oxygen and (ii) in terms of loss or gain of electrons.**

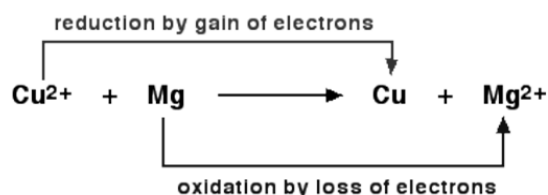
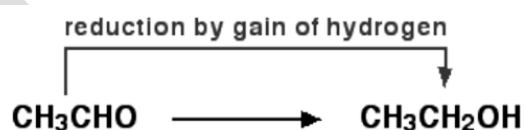
- (i) **In terms of loss or gain of oxygen:** When a substance **acquires oxygen** during a reaction, it is said to be oxidized (the process is called oxidation), and when a substance **relinquishes oxygen** during a reaction, it is said to be reduced (the process is called reduction).
- (ii) **In terms of loss or gain of electrons:** The species that **donates electrons** is identified as being oxidized, a process termed oxidation, whereas the species that **accepts electrons** is recognized as being reduced, a process termed reduction.

Oxidizing and Reducing Agents

- An oxidizing agent is defined as the substance that **accepts electrons** during a chemical reaction, thereby undergoing **reduction itself**.
- A reducing agent is defined as the substance that **donates electrons** during a chemical reaction, thereby undergoing **oxidation itself**.

Common examples of Oxidation

- Rancidity of Butter:** If butter is kept for a long time, its taste changes; this is an example of an oxidation reaction. As a result of oxidation, it gives a bad smell and taste.
- Rusting of Iron:** The reddish-brown powder that gets coated on objects made of iron is an example of an oxidation reaction.
- Souring of wine:** The wine becomes sour due to aerobic bacteria that allow the wine's alcohol content to oxidize.



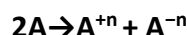
Redox Reactions

- Redox reactions are reactions in which the oxidation and reduction reactions occur simultaneously, such that **one substance is reduced and another is oxidized**.
- Redox reactions are common and vital to some of the basic functions of life, including photosynthesis, respiration, combustion, and corrosion or rusting.



Disproportionation Reactions

- In disproportionation reactions, a single substance can be both oxidized and reduced. These are known as disproportionation reactions, with the following general equation:



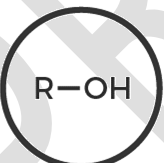
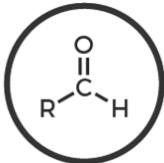
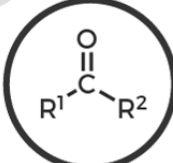
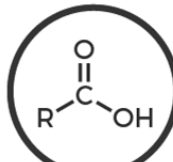
Where n is the number of electrons transferred.

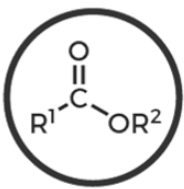
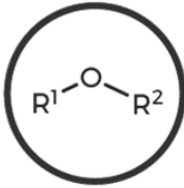


On the reactant side, H has an oxidation state of +1 and O has an oxidation state of -1, which changes to -2 for the product H_2O (oxygen is reduced), and the oxidation state is 0 for the product O_2 (oxygen is oxidized).

Functional Group Vs Homologous Series		
	Functional Group	Homologous Series
Definition	Functional groups are specific substituents within a molecule that are responsible for its characteristic chemical reactions.	A homologous series is a sequence of chemical compounds having the same functional groups and thus the same chemical reactivity.
Characteristics	Determines the chemical reactivity of a chemical compound.	Arrange different chemical compounds according to the similarity of chemical reactivity.
Reactivity	Different functional groups cause different chemical reactivity.	Have similar or nearly similar chemical reactivity among the members of the series.
Example	The functional group of carbohydrates is the aldehyde or ketone group.	Homologous series of primary alcohol starts with methane, ethane, propane, butane and pentane.

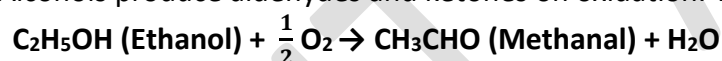
Basic Compounds

Class Of The Compound	Functional Group	Common Formula	Examples
Alcohol		R-OH	Ethanol, $\text{CH}_3\text{CH}_2\text{OH}$
Aldehyde		R-CHO	Ethanal, CH_3CHO
Ketone		R-CO-R	Acetone, CH_3COCH_3
Carboxylic Acid		R-COOH	Ethanoic acid (Acetic Acid), CH_3COOH

Ester		R-COO-R	Methyl Acetate, CH ₃ COOCH ₃
Ether		R-O-R	Dimethyl Ether, CH ₃ OCH ₃

Alcohol

- Alcohols are **colourless and sweet smelling**; **Liquid at room temperature**, although glycerol is a viscous liquid.
- Flammable** and produce a blue flame without any smoke; Alcohols are **acidic in nature**.
- They are **good solvents** and are used in medicines such as tinctures of iodine, cough syrups, and many tonics.
- It is the main ingredient of all alcoholic drinks.
- Oxidation of Alcohol**: Alcohols produce aldehydes and ketones on oxidation. The reaction is given below:



- The reaction of Alcohol with Metal**: Alcohol reacts with metal and forms alkoxides and releases hydrogen gas.

BCW Fact: Esterification is the process of combining a carboxylic acid (RCOOH) with an alcohol (ROH) to form an ester (RCOOR) and water.

A. Methanol

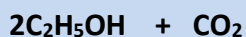
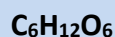
- Methanol or methyl alcohol or carbinol (CH₃OH) is a polar solvent with a hydroxyl group (-OH) attached to it.
- Also called **wood spirit** because it is **primarily produced by destructive distillation of wood**.
- It is a **colourless liquid, soluble in water** and forms explosive mixtures with air and burns with non-luminous flame.
- Methanol is a toxic alcohol; an overdose of methyl alcohol leads to metabolic acidosis, blindness and death.

Uses:

- Used in **radiators of automobiles as antifreeze**, in the preparation of perfumes, medicines, and dyes.
- Used in preparation for **methyated spirit or denatured spirit** (mixture of methyl alcohol & ethyl alcohol).

B. Ethanol

- Ethanol is a **colourless liquid** that has a boiling point of 351 Kelvin.
- Ethanol is also called ethyl alcohol. Its molecular formula is C₂H₅OH.
- Produced through the **fermentation of agricultural products** such as sugarcane, corn, wheat, etc.
- Zymase** catalyses the fermentation of glucose, converting it into alcohol (ethanol) and carbon dioxide.



Uses:

- Ethanol is a good fuel, and a mixture of petrol with ethanol is used as a fuel called **power alcohol**.
- Used as an **anti-knock additive** in petrol.
- Used in the **pharma industry** for the preparation of digestive syrups, cough syrup, etc.

Carboxylic Acid

- Carbon compounds containing a carboxyl functional group, **-COOH** are called carboxylic acids. The carboxyl group consists of a **carbonyl group attached to a hydroxyl group**.
- Carboxylic acids serve as starting material for several other important organic compounds such as anhydrides, esters, acid chlorides, amides, etc.

Ethanoic Acid

- It is also known as **Acetic acid (CH₃COOH)**.
- A 5-8% solution of acetic acid in water is called vinegar** and is widely used as a preservative in pickles.
- The melting point of pure ethanoic acid is 290 K and hence it often freezes during winter in cold climates. This gave rise to its name '**glacial acetic acid**'.
- Ethanoic acid reacts with absolute ethanol in the presence of an acid catalyst to give an ester.

Methanoic acid

- Methanoic acid** (Formic acid), along with histamine and acetylcholine, is present in the **leaves of Nettle** that cause painful stings.

Name	Molecular Formula	Structural Formula
Methanoic acid	HCOOH	$\begin{array}{c} \text{O} \\ \parallel \\ \text{H}-\text{C}-\text{O}-\text{H} \end{array}$
Ethanoic acid	CH ₃ COOH	$\begin{array}{c} \text{H} \quad \text{O} \\ \quad \parallel \\ \text{H}-\text{C}-\text{C}-\text{O}-\text{H} \\ \\ \text{H} \end{array}$
Propanoic acid	C ₂ H ₅ COOH	$\begin{array}{c} \text{H} \quad \text{H} \quad \text{O} \\ \quad \quad \parallel \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{O}-\text{H} \\ \quad \\ \text{H} \quad \text{H} \end{array}$
Butanoic acid	C ₃ H ₇ COOH	$\begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \quad \text{O} \\ \quad \quad \quad \parallel \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{C}-\text{O}-\text{H} \\ \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \end{array}$

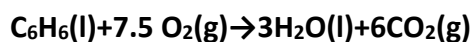
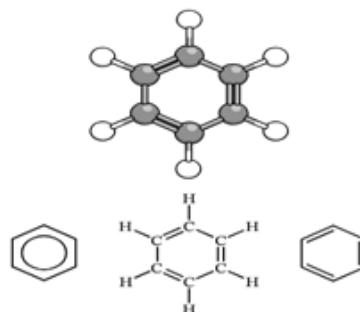
BCW Fact: While cutting an onion, we often get tears in our eyes because cutting the onion releases **sulfenic acid**, which activates the **lacrimal glands (tear-producing glands)** in our eyes.

Aromatic Hydrocarbon

- An aromatic hydrocarbon is a **hydrocarbon ring compound with alternating double and single bonds between carbon atoms forming rings.**
- Many of these compounds have a sweet scent, hence the term 'aromatic'. Aromatic hydrocarbons are found in almost every **petroleum mixture.**
- The simplest aromatic hydrocarbon is **benzene (C₆H₆).**

Benzene

- Benzene is an aromatic organic compound with molecular formula C₆H₆.
- It is a highly unsaturated compound.
- It has **twelve (12) Sigma bonds and three (3) Pi bonds.**
- When it is heated in the presence of oxygen, it burns with a sooty flame, producing carbon dioxide and water.



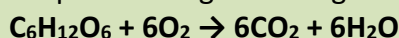
Sigma bonds are formed by **end-to-end** or **head-to-head** overlapping of atomic orbitals, whereas **Pi bonds** are formed by the **lateral** or **side-by-side** overlapping of atomic orbitals. **Sigma bonds are stronger than pi bonds.**

BCW Fact: **Sigma bonds** are formed by **end-to-end** or **head-to-head** overlapping of atomic orbitals, whereas **Pi bonds** are formed by the **lateral** or **side-by-side** overlapping of atomic orbitals. Sigma bonds are stronger than Pi bonds.

Hydrocarbon Fuel

- Hydrocarbon fuels are derived from **hydrocarbons, which include gasoline** and jet fuel.
- Unsaturated **hydrocarbons give a yellow flame** with a lot of black smoke, while saturated hydrocarbons give a clean flame.
- The colour of the flame is yellow** because of the **presence of unburnt carbon particles.**

Combustion is a chemical reaction in which a substance reacts quickly with oxygen and produces heat. The original substance is referred to as the fuel, and the source of oxygen is referred to as the oxidizer. E.g., The Combustion equation for glucose is given as follows;



Incomplete combustion is a type of combustion in which the fuel is not burned completely. This occurs when there is not enough oxygen present to allow the fuel to react completely to produce carbon dioxide and water. Instead, **carbon monoxide (CO)**, water (H₂O), and carbon dioxide are by-products of these reactions. For example, combustion in car engines, burning of coal, etc.

BCW Facts:

- Graphene**, a 2-dimensional allotrope of carbon, consists of a **single layer of carbon atoms** organized in a hexagonal lattice structure. Graphene is recognized as **one of the thinnest yet strongest materials** ever tested. This nanomaterial, Graphene, may be utilized in **solar cells, light panels, and touch screens** because it is a **transparent and good conductor**. Graphene was **discovered in 2004** by Andre Geim and Konstantin Novoselov, who were awarded the **Nobel Prize in Physics for their work in 2010.**
- Oxyacetylene welding**, often known as **gas welding**, is a technique that utilizes the combustion of oxygen and acetylene. When these gases are combined in the appropriate ratios within a handheld torch or blowpipe, they generate a flame that reaches a temperature of approximately 3,200 degrees Celsius.

6- Alloys

What Are Alloys?

- The **various properties of metals** can be **improved by mixing them with other metals**.
- A **homogeneous mixture** of two or more metals is called an alloy. It is prepared by mixing various metals in their **molten state in the required proportion** and then cooling the mixture to room temperature.
- **Stainless steel** is an alloy of Iron (Fe), Carbon (C), Chromium (Cr) and Nickel (Ni). It is very strong and does not rust. It is used in making cooking utensils, surgical equipment, dairy industry, etc.
- **Brass** is an alloy of Copper (Cu) and Zinc (Zn).
- **Bronze** is an alloy of Copper (Cu) and Tin (Sn). It is tough, resistant to corrosion, and is used to make statues, coins, medals, and utensils.
- **The properties of alloys are different** from the properties of their constituent metals. Like, the **electrical conductivity and melting point** of an alloy are less than those of pure metals.
 - For example, **Brass** (an alloy of Cu and Zn) and **Bronze** (an alloy of Cu and Sn) are **not good conductors of electricity**, whereas copper is used for making electrical circuits.
 - **Solder**, an alloy of lead and tin (Pb and Sn), **has a low melting point** and is used for **welding electrical wires** together.

BCW Fact: **Galvanization** refers to the method of applying a protective layer of zinc to steel or iron in order to safeguard these metals against rust and oxidation. This zinc coating can be achieved through four distinct techniques: Sherardizing, electrolytic plating, spraying (also known as schoop), and the hot dipping process.

Properties of Alloys

- **Alloys are stronger** than the metals from which they are made. They are **more resistant to corrosion**.
- They generally have **lower melting points**. They usually possess **lower electrical conductivity**.

Bell Metal

- Bell metal is an alloy **predominantly used in the manufacture of bells** and other similar instruments (thus the name).
- It is a type of bronze that contains a significant quantity of tin in its composition. Bell metal is an alloy of **copper and tin**:
- Typically, it consists of 78% Copper (Cu) and 22% Tin (Sn), in total.
- Cooking and dining utensils are also made from this alloy.

Bronze

- **Bronze is a copper and tin alloy**. Medals, coins, trophies, hefty gears and tools, and several types of electrical devices all employ it.
- It consists of 88% Copper (Cu) and up to 12% Tin (Sn), in total.
- Bronze has a far higher hardness than pure copper. It possesses better corrosion resistance.

Nichrome

- Nichrome is an alloy made **mostly of nickel and chromium**. It also contains iron and other elements as well.

- Nichrome contains 80-85% of Nickel (Ni); Chromium (Cr), Iron and other elements account for 15-20% of the total.
- Resistance wires** are the most common use of nichrome.

Steel

- It is largely an alloy of **iron and carbon**.
- Different varieties of steel are known to have various levels of carbon, as well as other elements (such as chromium, manganese, sulphur, phosphorus, nickel, copper, and molybdenum). Iron makes up the majority of steel's composition (it typically accounts for at least 75% of the alloy's weight).

Stainless steel

- Stainless steel** is made up of the following elements:
 - Iron (Fe)** content is between **85 and 88%**.
 - Chromium (Cr) content of at least 10.5% is required.
 - Carbon (C) and other alloying elements are less than 1.2%.
- Stainless steel** was invented by an English metallurgist, **Harry Brearley**, in 1913.
- Chromium protects steel** from **corrosion** and provides high-temperature resistance.
- By **adding nickel** to the alloy, stainless steel **becomes non-magnetic**.
- Carbon is mixed** to increase **hardness and strength**. An excess percentage of carbon increases brittleness and reduces weldability.

BCW Fact: An alloy of mercury metal with one or more metals is called an **amalgam**. The solution of sodium metal in liquid mercury is called sodium amalgam.

Name Of Alloy	Principle Metal	Composition	Properties	Uses
Duralumin	Aluminum	Al = 95% Cu = 4% Mn = 0.5%, Mg = 0.5%	Light weight and as strong as steel.	Aircraft frames, rockets, speed boats, automobiles.
Magnalium	Aluminum	Al = 95% Mg = 5%	Light weight, strong resists corrosion.	Used for making aeroplanes, household appliances, mirrors.
Stainless Steel	Iron	Fe = 83% C = 1% Cr = 15% Ni = 1%	Resists corrosion.	For making utensils, cutlery, surgical instruments.
Manganese Steel	Iron	Fe = 84% Mn = 15% C = 1%	Very tough and hard.	Used for making safes, armour rock cutters, etc.
Aluminum Bronze	Copper	Cu = 90% Al = 10%	Golden yellow in colour. Durable and resistant to corrosion.	Used for making artificial jewellery, coins, picture frames, etc.
Brass	Copper	Zn = 20-40% Cu = 60-80%	Lustrous, more malleable and ductile than copper.	Used for making shells of ammunition rounds, utensils, electric switches, statues, etc.
Bronze	Copper	Cu = 88% Sn = 12%	Hard but brittle, resistant to corrosion. Takes very high polish.	Used for making shells, statues, coins and utensils.

Gun Metal	Copper	Cu = 88% Sn = 10% Zn = 2%	Very hard, gives a very good cast.	Used for making barrels of guns, gears and bearings.
German Silver	Copper	Cu = 25-30% Zn = 25-30% Ni = 40-50%	As white as silver, malleable and ductile.	Used for making imitation silver jewellery, utensils, etc.
Bell Metal	Copper	Cu = 80% Sn = 20%	Hard and brittle, produces sonorous sound.	Used for making bells and gongs.

- **Oligodynamic effect:** It is the biocidal effect of metals, i.e., it kills microbes like bacteria, hence we see people using copper vessels to store water.
- **Corrosion effect:** With the passage of time, **Copper reacts with moist CO₂** and oxygen in the air and becomes **Green** due to the formation of "**basic copper carbonate**" which can be removed by using lemon or tamarind solution.
- **Silver becomes black** due to the formation of **silver sulphide, which** is another example of corrosion.

BCW Fact: Alloys like **Manganin, Constantan, and Nichrome** exhibit an extremely **low temperature coefficient of resistivity**, approximately 10^{-6} per degree Celsius ($^{\circ}\text{C}^{-1}$), coupled with a high resistivity. This characteristic makes these materials particularly suitable for the production of resistance wires and standard resistances. **Constantan** is an alloy of **Copper and Nickel**, while **Manganin** is an alloy of **Copper, Manganese and Nickel**.

7- Polymers, Bioplastics And Explosives

Polymers

- The word polymer is derived from two Greek words: "**Poly**→ many & **mer**→ unit."
- A **polymer** is a very large molecule formed by the combination of many small molecules. Polymers have **high molecular masses**, generally ranging from **10^3 to 10^7 atomic mass units**.
- The small molecules that combine to form a polymer are called **monomers**.

Polymerisation

- The process of forming polymers from monomers is called **polymerisation**.

2.	Protein	Amino acids	10.	Melamine	Melamine + Formaldehyde
3.	Starch, glycogen, cellulose	Glucose	11.	Terylene (Dacron)	Ethylene glycol + Terephthalic acid
4.	Natural Rubber	Isoprene	12.	Nylon 6-6	Adipic acid and hexamethylene diamine
5.	Synthetic Rubber (Neoprene)	Chloroprene	13.	Buna S	1,3-butadiene + Styrene
6.	PVC	Vinyl Chloride	14.	Buna N	1,3-butadiene + Acrylonitrile
7.	Polythene	Ethene	15.	Polyacrylonitrile	Acrylonitrile
8.	Teflon	Tetrafluoroethene	16.	Polystyrene	Styrene

Classification Of Polymers Based On Molecular Forces

Polymers are categorized into four types based on the intermolecular forces among the monomer molecules. These are - 1. Fibres, 2. Elastomers (Rubber), 3. Thermoplastics, 4. Thermosetting.

Fibres

- A very thin, thread-like strand, characterized by its significant tensile strength and high modulus from which cloth is made, is called a **fibre**. Mainly two types of fibres:

A. Natural fibres

- The fibres obtained from plants and animals are called natural fibres. e.g., Cotton, silk, wool, jute, flax

B. Synthetic fibres

- The manmade fibres produced from chemical substances are called synthetic fibres. e.g., Rayon, Nylon, Polyester, Acrylic, Kevlar

Synthetic Fibres	Uses
Rayon	- Often referred to as artificial silk, Rayon is the first man-made (artificial) fibre. - Obtained by the chemical treatment of wood pulp, which contains cellulose. Often blended with cotton to make bed sheets or combined with wool to make carpets.
Nylon	Nylon is a polyamide. It was the first fully synthetic fibre and it was initially made from water, coal, and air. - Nylon is strong, light, elastic, lustrous and easy to wash.
Polyester	- Polyester is made up of many units of an ester. These are the products formed through the polycondensation of dicarboxylic acids and diols. - Suitable for making dress material. Terylene is a popular type of polyester.
Acrylic	Acrylic is cheaper and more durable than natural wool, so it is often used as a substitute for wool. Acrylic fibers exhibit excellent resistance to stains, chemicals, insects, and fungi.

Kevlar	Kevlar is a material commonly used to make bulletproof vests. Kevlar is classified as an aramid fibre, which is an abbreviation for 'aromatic polyamide'. Kevlar was developed by Stephanie Kwolek, a chemist, in 1965.
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Natural Rubber And Vulcanization Of Rubber

Natural Rubber

- It is produced from **latex, a milky colloid drawn from the rubber tree** (*Hevea brasiliensis*) or some other plants. When coagulated, latex transforms into a **soft, plastic and sticky substance (crude rubber)**, which is then vulcanized (cured).
- Natural rubber is composed of **poly-isoprene molecules**.
- It is an **elastomer, sensitive to temperature and behaves as a thermoplastic**. When **vulcanization is performed**, it improves the characteristics of natural rubber and increases its application.

BCW Fact: Rubber latex is derived from the **bark of the rubber tree**, which is obtained in countries such as India, Sri Lanka, Indonesia, Malaysia, and South America.

Disadvantages of raw natural rubber

- It becomes **soft at elevated temperatures (above 335 K)** and **brittle at lower temperatures (below 283 K)** and, it shows high water absorption capacity.
- Additionally, its **tensile strength is limited to 200 Kg/cm²**.

Vulcanization of Rubber

- It is the process of improving the elasticity and strength of rubber by **heating it in the presence of sulphur at a temperature of 373-415 K to produce vulcanized rubber**.
- It results in **three-dimensional cross-linking of the chain rubber molecules (poly-isoprene)** bonded to each other by sulphur atoms.
- Vulcanized rubber doesn't possess temperature- sensitive properties of softness or brittleness like natural rubber. Once the rubber is vulcanized, it becomes a **thermoset material**.
- In the **production of tyre rubber**, a cross-linking agent comprising **5% sulphur** is used. The Vulcanization process was **invented by Charles Goodyear in 1839**.

Properties of rubber improved by vulcanization: High tensile strength; high elasticity; improved hardness; high wear and tear resistance; abrasion resistance; and resistance to solvents.

Synthetic rubber

- These are **Elastomers whose elastic properties are similar to those of natural rubber, e.g., Buna-S, Buna-N, Neoprene**.
 - Synthetic rubber may have chemical composition similar to that of natural rubber (synthetic poly-isoprene). Other kinds of synthetic rubbers are: **polybutadiene, poly(styrene-butadiene-styrene), polychloroprene (Neoprene), polyisobutylene (Butyl rubber), and silicone**.
- A. Neoprene:** Neoprene, a type of synthetic rubber, is produced through the free radical polymerization of **chloroprene**. It exhibits exceptional **resistance to both vegetable and mineral oils** and is used in the **manufacturing of conveyor belts, gaskets, and hoses**.

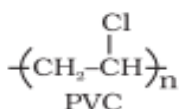
- B. Buna-N:** Buna-N, or Nitrile Butadiene Rubber (NBR), is a synthetic copolymer formed from acrylonitrile (ACN) and butadiene monomers. This material exhibits **resistance to petrol, lubricating oils, and various organic solvents**. It is commonly utilized in the **production of oil seals, tank linings**, etc.

Plastic

- Plastics are also polymers. Plastics are durable, non-reactive, and poor conductors. These are mainly two types: **Thermoplastic** and **Thermosetting plastics**.

Thermoplastic

- A plastic that can be softened repeatedly by heating and can be **moulded into different shapes multiple times** is called a **thermoplastic**. These are long-chain molecules that are either linear or slightly branched.
- Examples:** Polythene, PVC, Polycarbonates, Polystyrene, and Teflon.



Thermosetting plastics

- A plastic that, once set, **does not soften upon heating and cannot be moulded a second time** is called a thermosetting plastic.
- Examples:** Bakelite and Melamine.



Bio-Degradable Polymers

- They are **designed and developed to address the problem of polymeric solid wastes**.
- These polymers contain functional groups similar to the functional groups present in biopolymers.
- Examples:** Polylactic Acid (PLA), Polyhydroxyalkanoates (PHA), Polyhydroxybutyrate (PHB), Polyglycolic Acid (PGA), Polycaprolactone (PCL), and Starch-Based Plastics.

A. PHBV (poly beta-hydroxybutyrate-co-beta-hydroxy valerate)

- It is obtained by the **co-polymerization** (more than one monomeric species is allowed to polymerise) of 3-hydroxybutanoic acid and 3-hydroxypentanoic acid.
- Used in speciality **packaging, orthopedic devices** and in controlled release of drugs.

B. Nylon 2-Nylon-6

- It is an alternating copolymer of glycine ($\text{H}_2\text{N}-\text{CH}_2-\text{COOH}$) and amino caproic acid [$\text{H}_2\text{N}(\text{CH}_2)_5\text{COOH}$].

Harmful Effects of Biodegradable Polymers:

Large amounts of **biodegradable waste** can harm the environment.

- Their decomposition promotes the growth of **microorganisms** that may spread **infectious diseases** to humans, plants, and animals.
- Some biodegradable polymers may not completely decompose, leading to **microplastic pollution**.
- Their disposal produces **unpleasant odour** and may release **greenhouse gases** during incineration.

Non-Biodegradable Polymers

- Non-biodegradable polymers are **those that do not undergo significant degradation or decomposition by natural biological processes within a reasonable timeframe.**
- Examples:** Glass, batteries, plastic bottles, tetra packs, medical waste, and carbon paper.

A. Polystyrene

- Polystyrene is a **synthetic polymer** composed of the monomer of the aromatic hydrocarbon styrene.
- Tough to disintegrate and provide both solid and long-lasting commodities.
- Polystyrene serves multiple purposes, including its application as an insulator, a packaging material, and in the manufacturing of toys.

B. Polythene

They are mainly of three types:

- High-density Polythene (HDP):** The formation of this type of polythene occurs through the process of **addition polymerisation of ethene (ethylene)** in a hydrocarbon solvent, utilizing a catalyst such as triethylaluminium and titanium tetrachloride (Ziegler-Natta catalyst) at temperatures ranging from 333 K to 343 K and under pressures of 6 to 7 atmospheres
- Low-density Polythene (LDP):** It is obtained by the polymerisation of ethene under high pressure (1000–2000 atmospheres) and temperature (350 K to 570 K) with traces of dioxygen or a peroxide initiator (catalyst). It is **chemically inert** and a poor conductor of electricity.
- Ultra-high molecular weight polyethylene:** It is a superior type of polyethylene, characterized by its remarkably lengthy polymer chains.

Harmful Effects of Non-Biodegradable Polymers:

They harm the environment in various ways.

- Non-biodegradable plastic containers** in the oceans and estuaries can harm **fish, seabirds and other marine life.**
- They can cause soil to become more **acidic or alkaline**, and reduce fertility.
- At each trophic level, toxic compounds like non-biodegradable polymers, microplastics, and DDT accumulate. Since humans are at the top of the food chain, their bodies contain the highest concentration of these substances. This phenomenon is called **“biological magnification”**.

Polymers of Commercial Significance

Polymer	Monomer	Uses
Polypropene	Propene	Ropes, toys, pipes, fibres, etc.
Polystyrene	Styrene	Insulator, wrapping material, toys, radio and television cabinet.
Polyvinyl Chloride (PVC)	Vinyl Chloride	Rain coats, hand bags, vinyl flooring, water pipes.
Urea-Formaldehyde Resin	Urea Formaldehyde	Unbreakable cups and laminated sheets.
Glyptal	Ethylene Glycol Phthalic Acid	Paints and lacquers

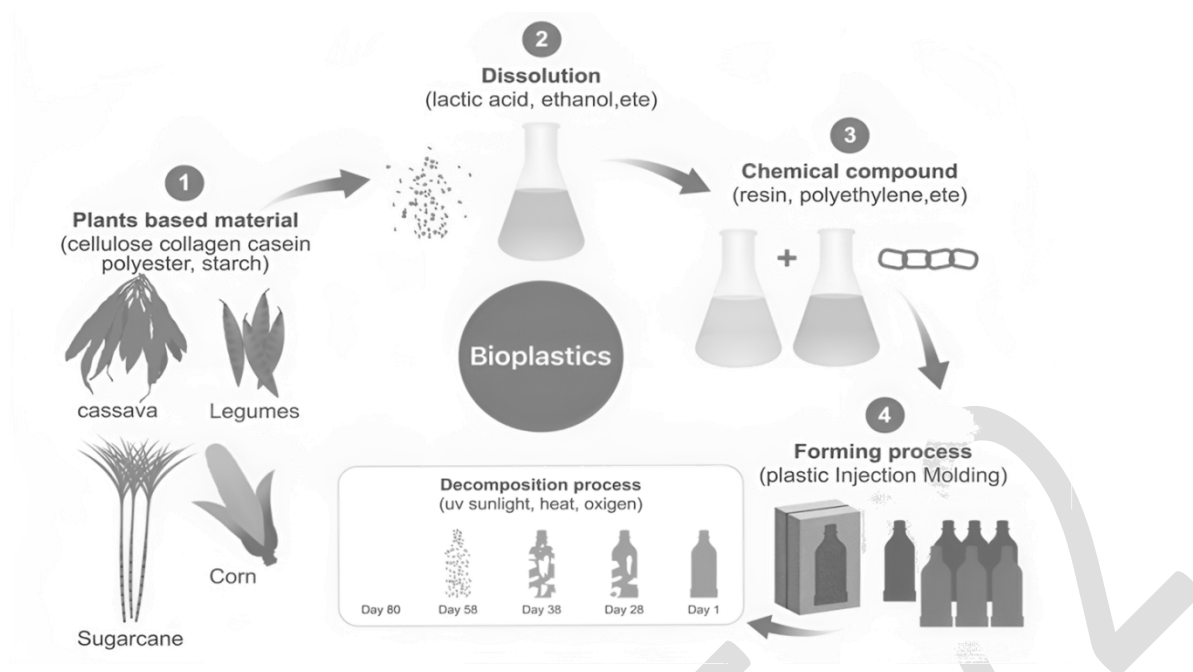
Bakelite	Phenol Formaldehyde	Combs, electrical switches, handle of utensils and computer discs.
Kevlar	Polyamides	Bullet proof jackets and light weight bullet proof helmets.
Natural Rubber	Isoprene	Medical devices, surgical gloves, tires, etc.
Polyethylene terephthalate	Ethylene glycol and Terephthalic acid	Packaging of food and beverages, particularly single-serving soft drinks, juices, and water.
Teflon	Tetrafluoroethylene	Non-stick pan
Nylon 6	Caprolactam	To make clothes, safety nets and ropes.

Teflon is the registered trade name of the highly useful plastic material **polytetrafluoroethylene (PTFE)**. Teflon is classified as a **solid fluorocarbon**, being a polymer composed exclusively of carbon and fluorine atoms.

- PTFE is classified as a thermoplastic because its melt-viscosity is so high that it will only melt and soften under extreme temperatures.
- Teflon is produced by heating **tetrafluoroethylene** under high pressure in the presence of a free radical or **persulfate catalyst**. This substance is **chemically inert and resistant to corrosive agent attacks**. It is commonly utilized in the fabrication of oil seals and gaskets.
- It is a linear polymer chain, and a **melt-transition point can be detected at 327 °C (620.6 °F)**
- It is mainly used for making non-sticky pans, as it has **"Fluorine" molecules that repel almost all materials, thus preventing them from sticking.**

Bio Plastics

- Bio-based plastics are **developed from renewable biomass (plants) such as corn, sugarcane, vegetable oil or wood pulp**. Some bioplastics are **biodegradable**. Biodegradable plastics are those which possess the characteristics of biodegradability and composability.
- They can be converted into natural substances **and degraded by the action of micro-organisms in the environment**.
- Bioplastics typically have a **lower melting point in water**, are significantly less harmful.
- Bioplastics comes **from renewable sources and can be used to reduce the problem of hazardous plastic waste**.



Types of Bioplastics

Bioplastics can be prepared from a variety of **materials like starch, sugar, cellulose**, etc. There are different types of bioplastics depending on the source from which they are obtained. They are:

- Cellulose-based Bioplastics:** Usually produced from **wood pulp** and are used to make film-based products such as wrappers and seal to maintain freshness in prepared food.
- Starch-based Bioplastics:** These are derived from **corn starch** and are often mixed with biodegradable polyester.
- Protein-based Bioplastics:** These are derived from plant proteins such as soya protein, gluten, and gelatine.
- Aliphatic Polyester Bioplastics:** Natural bacterial fermentation of cost-effective carbon sources, including molasses, sucrose, lactose, glycerol, oils, and methane, primarily produces aliphatic bioplastics. E.g., Polylactic acid (PLA) and Polyhydroxyalkanoates (PHA).
- Polyamide 11 Bioplastics:** These are derived from castor oil.

Explosives

- Explosive materials consist of chemical compounds or mixtures that can rapidly decompose, resulting in the sudden and violent release of heat, gas, and pressure. These materials find applications in numerous sectors, such as the military, mining, construction, and pyrotechnics.

Dynamite

- Dynamite is an explosive made of **nitro-glycerine**. It was **invented by the Swedish chemist and engineer Alfred Nobel** in Gaeltacht in 1866 and was patented in 1867. Today, dynamite is mainly used in mining.

T.N.T. (Trinitrotoluene)

- It is a yellow solid which is formed by **stepwise nitration of Toulene**.
- Formula - $\text{C}_6\text{H}_2(\text{NO}_2)_3\text{CH}_3$
- Chemical name : **Trinitrotoluene, or 2-methyl-1, 3, 5-trinitrobenzene**
- It is used as explosive. It was first prepared by **German chemist Joseph Wilbrand**.
- It is relatively **insensitive to spark and needs detonator for explosive** i.e., its **explosion can be controlled**. **Hence, used in demolition**. When **mixed with Ammonium Nitrate (NH_4NO_3)** it forms another explosive substance called **Amatol**.

T.N.P (Trinitrophenol)

- It is also called **picric acid**. The compound is represented by the formula $(\text{NO}_2)_3\text{C}_6\text{H}_2\text{OH}$. Its IUPAC name is **2, 4, 6-trinitrophenol (TNP)**.
- It is primarily utilized as an explosive. Additionally, it has found applications in the medical field, serving as an antiseptic and in the treatment of burns, as well as in the production of dyes

T. N. G. (Trinitroglycerin)

- It is also called **Nobel Oil** or nitroglycerin or glyceryl trinitrate or 1,2,3-trinitroxypropane. Nitroglycerin is an ester. It was discovered by **Ascanio Sombroso in 1846** and it is **colourless in its purest form**. It is used in **manufacture of dynamite**.

R. D. X. (Royal Demolition Explosive or Research Department Explosive)

- **Chemical name**-1,3,5-trinitroperhydro—1,3,5-triazine
- **Other name**- Cyclotrimethylene trinitramine
- It was discovered by **George Friedrich Henning of Germany**.
- **Chemical formula** - $\text{C}_3\text{H}_6\text{N}_6\text{O}_6$

BCW Facts:

- **PET (Polyethylene Terephthalate)** is a type of **polyester** widely used in manufacturing plastic bottles, food containers, packaging materials, fabrics, films, wires and even engineering applications.
- **Melamine** is a **thermosetting plastic** known for its fire resistance and superior heat tolerance compared to other plastics. It is commonly used in the production of **floor tiles, kitchenware, unbreakable crockery and fire-resistant fabrics**.
- **Gunpowder** has been extensively used as a propellant in various applications, including **firearms, artillery, rocketry, and pyrotechnics**. This substance is a mixture of **sulphur, potassium nitrate (commonly known as saltpetre), and charcoal**.

8- Food Preservative

Food Preservation

- Food provides energy to all of us. However, Food that is unhealthy or spoiled is the leading cause of disease in humans. This is referred to as food poisoning. To avoid this, food must be preserved in a variety of ways.
- **Food preservation** is one of the methods of protecting food from **microbial growth**. The main idea behind it is to **slow down the metabolism, preventing decay and increasing the shelf life**.
- Food preservation helps in food storage by **increasing the shelf life of food**, and it minimizes food wastage.

Advantages of food preservation

- **Germ and microorganisms do not grow easily in preserved food**.
- Preservation enables us to enjoy seasonal fruits like strawberries and mangoes even during the off-season.
- The **nutritional content of food can be improved or maintained** by adding certain additions to the recipe. For example, by adding vitamin C to bread.

Disadvantages of food preservation

- **Excess use of salt and sugar** in food preservation may be harmful to **health**.
- Some methods of **food preservation** may lead to **loss of nutrients**.
- Certain preservatives or chemicals may also cause **allergic reactions or other health problems** in sensitive individuals.

Methods used for Food Preservation

- **Removal of Microorganisms:** Drying, salting, etc.
- **Decreasing Temperature:** Freezing.
- **Increasing Temperature:** Pasteurization.
- **Reduced Water Availability:** Drying, vacuum, lyophilization.
- **Chemical-Based Preservation:** Addition of specific inhibitory compounds (Example: organic acids, nitrates)
- **Radiation:** Use of **ionizing (gamma rays)** and **non-ionizing (UV) radiation**.

Drying

- It is an ancient practice of food preservation. This method lowers the water activity, thereby **inhibiting the growth of bacteria**.
- Drying under **sunlight and wind decreases the moisture content**.

Freezing

- Freezing is the method of keeping ready-made food in a cold environment. **Freezing slows down the rate of biochemical reaction (i.e., metabolism), thereby increasing the shelf life.**

Vacuum Packing

- A vacuum or airless environment is created by baggage and bottling in airtight containers. Since there's no **chemical element within the created vacuum, microorganisms gradually die.** It is sometimes used for preserving dry fruit.

Salting and Pickling

- With the use of a good quantity of salt and spices, the moisture comes out from the body of the microorganism through osmosis.
- Pickling suggests protecting food in brine (salt solution) or marinating in vinegar (acetic acid). Salt kills and inhibits the expansion of microorganisms.**
- In industrial pickles, **sodium benzoate is added to increase shelf life or extend the time period.**

Sugar syrup

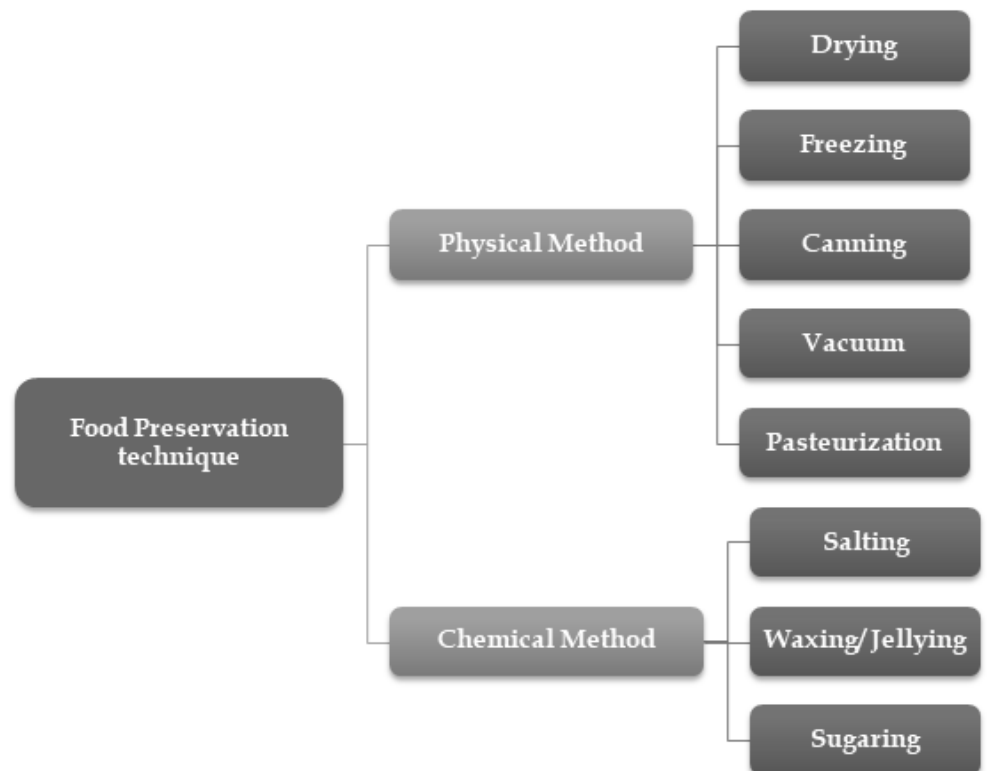
- Sugar is used in the form of **syrup** to preserve fruits, or in **crystallised form** when the substance to be preserved is cooked in sugar until crystallisation occurs, as in **candied peel and ginger.**
- Some fruits are also preserved by giving them a **superficial coating of sugar syrup.**

Pasteurization

- Pasteurization is a simple process in which mild heat treatment is given for various packaged and non-packaged foods (such as milk and fruit juice) that **kills most microorganisms and then instantly cools down.**

Organic Acid and Salts

- Some organic acids and salts such as **Sorbic Acid / sulphides or Benzoate / Benzoic acid or Nitrates / nitrites** are used in small prescribed amount to prevent microbial growth in food items.



- **Sodium benzoate (C_6H_5COONa)**, a white colour salt, is extensively used for the **preservation of food items** such as **fruit juices, soft drinks, squashes, jams and pickles**.
- **Nitrates/nitrites** are used for the preservation of meat products.

Canning and bottling

- **Bottling and canning** involve heating food under pressure and sealing it in **airtight containers**. The food is heated to kill microorganisms and then sealed to prevent further contamination. Once opened, the food becomes susceptible to **spoilage again**.

Waxing

- **Wax coating on fruits** plugs the **pores** and decreases **metabolism**, minimizing interaction with the external environment and thereby **increasing shelf life**.
- Some common waxes used include **bee wax, sugarcane wax**, and **other permitted waxes**.

Jellying

- It is a form of food preservation where the food to be **preserved is cooked in a substance that forms a natural gel, thereby inhibiting the growth of microbes**.
- Fruits are typically preserved as jellies, jams, or preserves using natural gelling agents like **cellulose** and **sugar**.

BCW Fact: Fruits with **high Pectin (Jelly making content)** are preferred for **jelly making**. **Guava** has high Pectin content and is therefore well suited for making jelly.

Radiation

- **Ultraviolet radiation** is used to **reduce surface contamination in various foods**. This short-wavelength light is used in the cold storage units of meat processing plants.

Lyophilization

- Lyophilization is also known as **Freeze-drying**. It is used in the biotechnology and biomedical industries to preserve vaccines, blood samples, purified proteins and other biological material. The basic methodology of lyophilization is to **remove the water or wet content** from the food.

BCW Facts:

- **Rancidity** – The process by which **fats and oils get oxidized**.
- Chips packets are filled with Nitrogen gas to prevent **rancidity**.
- **Cinnamon and Clove** contain **eugenol**, which has preservative properties.

Food Preservatives

- Among the most frequently used preservatives are **table salt, sugar, vinegar (or acetic acid), vegetable oils, and sodium benzoate**. Additionally, salts of **Sorbic acid** and **Propanoic acid** are employed as preservatives.

Antioxidants In Food

- These substances contribute to the preservation of food by **slowing down the effects of oxygen** on it.
- The two most commonly recognized antioxidants are **butylated hydroxy toluene (BHT)** and **butylated hydroxy anisole (BHA)**. Incorporating BHA into butter can extend its shelf life from several months to several years.

BCW Fact: Monosodium Glutamate (MSG) is a **flavour enhancer** derived from the sodium salt of glutamic acid, an amino acid present in numerous foods as well as in the human body. It is also known as **Chinese salt**. Commonly recognized by its brand name, **Ajinomoto**, MSG is widely used in diverse culinary practices, particularly in Asian cuisine. It is frequently added to processed foods, soups, sauces, snacks, and ready-to-eat meals. MSG provides an **umami flavour**, characterized as savoury or meaty, which is regarded as the fifth basic taste alongside **sweet, bitter, sour, and salty**.

9- Periodic Table With Important Elements

The **Periodic Table** is one of the most important tools in **chemistry**, providing a **systematic classification of elements** based on their **atomic number, electron configuration, and recurring chemical properties**. It helps scientists and students understand **element behaviour, chemical reactions, and relationships among elements**. The idea of classifying elements began in the **19th century**, and the modern periodic table was developed by Dmitri Mendeleev in **1869**. It has since been modified with the discovery of new elements and advances in **atomic theory**.

Elements

- An **element** is the simplest form of a **chemical substance** that cannot be broken down into simpler substances by **chemical reactions**.
- In **1800**, only **31 elements** were known, which increased to **63 by 1865**. So far, scientists have discovered **118 elements**.
- Each element has unique **physical and chemical properties**, but some elements show **similar behaviour and properties**. Therefore, arranging elements systematically makes their study easier.
- The **Periodic Table** groups elements with similar properties together. Many scientists proposed different classifications in the past, leading to the development of the **modern periodic table**.

Dobereiner's Law Of Triads

- In **1817**, the German chemist Johann Wolfgang Dobereiner attempted to classify elements with similar properties into groups. By **1829**, he identified groups of **three elements** with similar properties, called **triads**.
- According to **Dobereiner's Law of Triads**, when elements are arranged in **increasing order of atomic weight**, the **atomic weight of the middle element** is approximately the **arithmetic mean of the other two elements**.

Category	Name Of The Element	Atomic Mass
A	Li	7
	Na	23
	K	39
B	N	14
	P	31
	As	74
C	B	10.8

	Al	27
	Ga	69.7

Limitations

- It did not apply to the elements of very low or high atomic masses.
- Many new elements were discovered in the 18th and 19th centuries, which made the classification of elements in Dobereiner's triads difficult and also impossible in some cases.

Newland Octave Series

- The English chemist John Alexander Newlands proposed the **Law of Octaves** in **1865**. He arranged elements in **increasing order of atomic weight** and observed that the **properties of every eighth element were similar**, just like the repetition of musical notes after the seventh note in "Sa Re Ga Ma Pa Dha Ni," where after the 7th note (Ni), the 1st note (Sa) repeats itself.
- Newlands arranged about **56 known elements** according to this pattern in his **octave series**.

Limitations

- The issue with his classification was that he placed **cobalt and nickel in the same position**.
- The position of **inert gases was not defined**.
- His **rules failed after Calcium**.

Mendeleev's Periodic Table

- Although Johann Wolfgang Döbereiner began the study of periodic relationships, it was Dmitri Mendeleev who first published the **Periodic Law** in **1869**. It states that: "**The properties of elements are a periodic function of their atomic weights.**"
- Mendeleev arranged elements in **horizontal rows (periods)** and **vertical columns (groups)** according to their **increasing atomic weights**. Elements with **similar properties** were placed in the same group. Hence, Mendeleev is known as the **Father of the Modern Periodic Table**.
- Mendeleev's table was a major improvement over **Dobereiner's triads** and helped in the understanding of **element valency**. For example, elements of **Group 1** such as **hydrogen, lithium, sodium, potassium, rubidium, and caesium** show a **valency of 1**.

Limitations of Mendeleev's Periodic Table

- Elements with very different properties, such as **copper and silver**, were placed in the same group as **alkali metals**.
- The position of **hydrogen** could not be fixed properly.
- **Isotopes** were not given separate positions.
- Mendeleev could not explain the **cause of periodicity** or the proper separation of similar elements.
- Although he correctly predicted undiscovered elements like **gallium** and **germanium**, the table still contained **gaps and inconsistencies**.

Modern Periodic Table

- In **1913**, the English physicist Henry Moseley studied the **X-ray spectra** of elements and concluded that the **atomic number** is a more fundamental property than atomic mass. This led to the modification of Mendeleev's law into the **Modern Periodic Law**.

Modern Periodic Law

- It states that: "**The physical and chemical properties of elements are periodic functions of their atomic numbers.**"
- The **Modern Periodic Table** arranges elements in **increasing order of atomic number**. It contains **18 groups** (vertical columns) and **7 periods** (horizontal rows). Elements in the same group have **similar chemical**

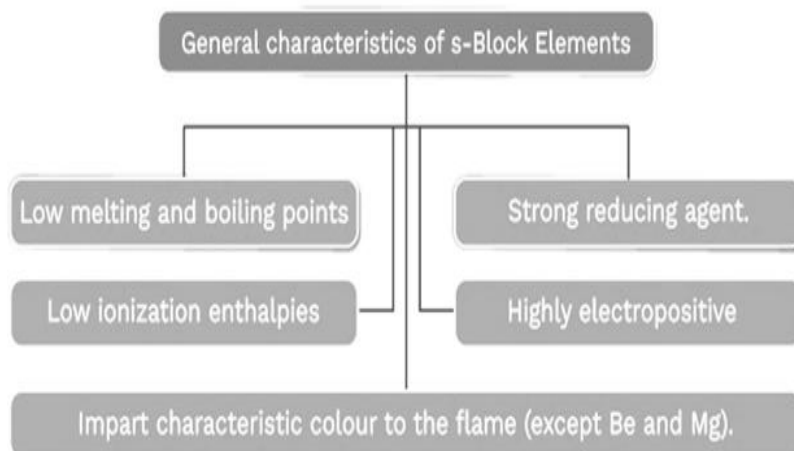
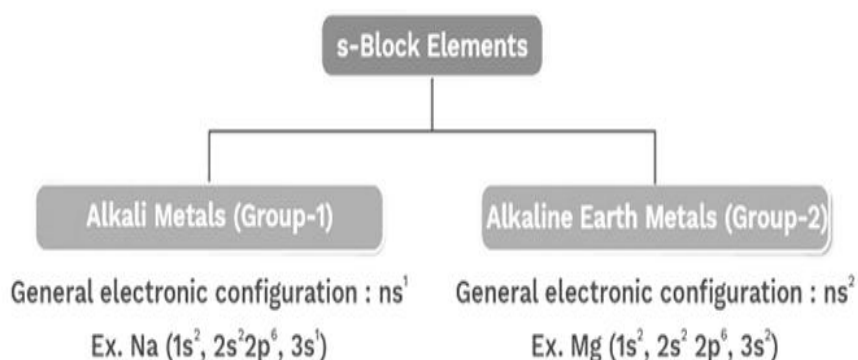
properties. Based on electronic configuration, elements are divided into **s-block, p-block, d-block, and f-block.**

- The periodic table has **seven periods**, and the period number corresponds to the **highest principal quantum number (n)** of the elements in that period. The first period contains **2 elements**, while the next periods contain **8, 8, 18, 18, and 32 elements**, respectively.
- The **seventh period is incomplete** and, like the sixth period, can theoretically contain **32 elements**.
- The **14 lanthanoids and 14 actinoids** of the sixth and seventh periods are placed separately at the bottom of the table.

BCW Fact: Mendelevium (Md) is a synthetic element that holds the **atomic number 101** and is classified within the actinide series of the periodic table. The element is named in honour of **Dmitri Mendeleev**.

S - Block Elements

- The elements classified as **Group 1 (alkali metals)** and **Group 2 (alkaline earth metals)**, characterized by their outermost electronic configurations of ns^1 and ns^2 , are categorized as **s-Block Elements**.
- They all are reactive metals with low **ionization enthalpies**, hence **low electronegativity**.
- They lose the outermost electron(s) readily to form **+1 ions** (in the case of alkali metals) or **+2 ions** (in the case of alkaline earth metals). Hence, they form **an ionic bond** with fixed oxidation states.
- Their **metallic character and reactivity** increase as we go down the group. Due to their **high reactivity**, they are **not found in the free state in nature**.
- The **oxides and hydroxides** of s-block elements are **basic in nature**.
- Alkaline earth metals** show chemical properties similar to **alkali metals**, but differences arise due to their **smaller atomic and ionic sizes** and **higher cationic charge**. Therefore, their **oxides and hydroxides are less basic** than those of alkali metals.



P - Block Elements

- The **p-block elements** are found in **Groups 13–18** and have outer electronic configuration from **ns^2np^1 to ns^2np^6** .
- Most p-block elements are **non-metals and metalloids**, though some are **metals**.

- At the **end of each period**, one can find a **noble gas element**.
- **Noble gases (Group 18)** have completely filled valence shells (ns^2np^6), making them **chemically less reactive**.
- **Halogens (Group 17)** and **chalcogens (Group 16)** have high **electron gain enthalpy** and readily gain electrons to attain **noble gas configuration**.
- The **non-metallic character increases across a period**, while the **metallic character increases down a group**.

- **Helium, Neon, Argon, Krypton, Xenon, and Radon** belong to **Group 18** of the periodic table. (**Noble gases or inert gases**).
- Among them, **Radon** is the only **radioactive noble gas** and is not found freely in the atmosphere.
- **Argon** is used in **incandescent electric bulbs**, while **Neon** is used in **advertising and beacon lights**.
- Air in oxygen cylinders used for diving is mixed with **helium** because helium is **less soluble in blood than nitrogen**.
- Noble gases exist as **monatomic gases** and are **colourless, odourless, and tasteless**. Their solubility in water is minimal. They have **very low melting and boiling points** due to weak **dispersion forces**.
- **Helium** has the **lowest boiling point** of any known substance, about **4.2 K**.

BCW Fact: Most non-metals exist as solids or gases at room temperature, except **Bromine**, which exists in a **liquid state**. Bromine is used in fire extinguishers and pesticides. Inhalation of bromine vapours can have toxic effects on the human body. Bromine is a **reddish-brown liquid**.

D - Block Elements

- **d-block elements** are located between the **s-block and p-block elements** in **Groups 3–12** of the periodic table. Their outermost shell contains **1 or 2 electrons in the ns orbital**, while the last electron enters the **(n–1) d orbital**.
- All d-block elements are **metals**. They commonly show **coloured ions, variable oxidation states, paramagnetism, and catalytic properties**.
- Their properties are **intermediate between s-block and p-block elements**.
- Since they represent the transition from **highly electropositive s-block elements** to **less electropositive p-block elements**, they are called **transition elements**.
- Transition elements have a **partially filled d-subshell** in their atomic or ionic state.

F - Block Elements

- The f-block consists of two series of **inner transition elements**: the **lanthanoids** (the fourteen elements from Cerium (Ce, Z = 58) to Lutetium (Lu, Z = 71)) and the **actinoids** (the fourteen elements from Thorium (Th, Z = 90) to Lawrencium (Lr, Z = 103)). These elements are characterized by the filling of their 4f and 5f orbitals, respectively.

NOTE: We know that the lanthanoids follow lanthanum, and the actinoids follow actinium. Lanthanum (La, Z = 57) and Actinium (Ac, Z = 89) are not f-block elements because their differentiating electrons occupy d-orbitals instead of f-orbitals. **Lanthanum (La)** and **Actinium (Ac)** are technically part of the **d-block elements** in the periodic table, despite their common association with the f-block (lanthanoids and actinoids).

Periodicity In Properties

Atomic Radius

There are three types of atomic radii: covalent radius, ionic radius and metallic radius.

- **Covalent radius:** Half the distance between the nuclei of two covalently bonded atoms of the same element.
- **Ionic radius:** The radius of an atom that forms an ionic bond.
- **Metallic radius:** The radius of an atom bonded to another atom through metallic bonding.

Trend of Atomic Radii across the Periodic Table:

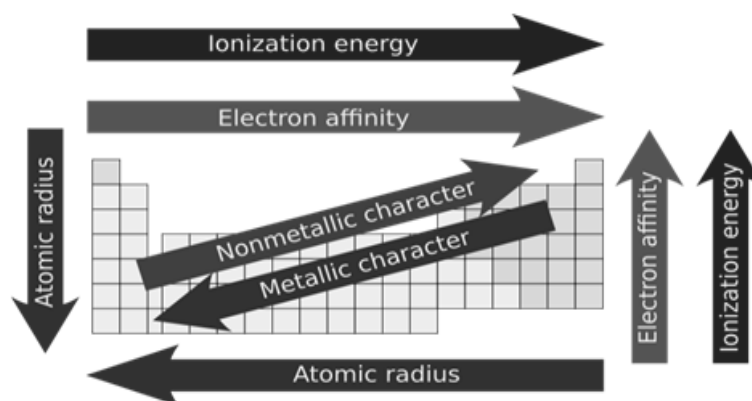
- **Atomic radius decreases across a period** because the **effective nuclear charge increases**, pulling electrons closer to the nucleus.
- Within a family or a group in a periodic table, the **atomic radius increases regularly with atomic number**.

BCW Facts:

- **Cations are smaller** than their parent atoms.
- Among **isoelectronic species**, the species with the **greater positive nuclear charge** has the **smaller radius**.

Trends In Physical Properties

Various physical properties of elements, including melting and boiling points, heats of fusion and vaporization, and atomization energy, exhibit periodic variations.



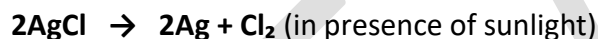
Property	Definition	Across The Period (→)	Down The Group (↓)
Ionization enthalpy/ Ionization energy	Amount of energy required to remove electrons from the outermost or valence shell of an isolated gaseous atom.	Increases	Decreases
Electron gain enthalpy	Amount of energy released in the process of adding an electron to an isolated gaseous atom. Therefore, it is negative.	Increases	Decreases
Electronegativity	Tendency to attract shared pairs of electrons.	Increases	Decreases

Atomic radii	Distance between the outermost shell and the nucleus of the atom.	Decreases	Increases
Metallic character	Tendency to lose an electron.	Decreases	Increases

Metals

More than 75% of the known elements have the characteristic properties of metals.

- The ability of metals to be drawn into **thin wires** is called **ductility**. **Gold** is the most ductile metal. Metals that produce a **ringing sound** when struck are called **sonorous**.
- Metals are **good conductors of heat and electricity**, e.g., **copper and silver**.
- Almost all metals react with **oxygen** to form **metal oxides**, though their reactivity **towards oxygen differs**.
- Example: Potassium and sodium** react vigorously with oxygen and may catch fire **if kept in the open**; hence, they are kept immersed in **kerosene oil**.
- At ordinary temperature, metals like **magnesium, aluminium, zinc, and lead** form a thin **oxide layer** on their surface, which protects them from further oxidation.
- When silver chloride is exposed to sunlight, it turns grey due to the decomposition reaction that forms Silver metal (grey in colour) and liberates chlorine gas.



The above reactions are used in black and white photography. To prevent this decomposition by sunlight, **silver chloride is kept in dark bottles**.

Heavy and Light Metal

- Heavy metals** are metals with relatively **high density, atomic weight, or atomic number**, while metals with low density are called **light metals**. **Lithium** is the **lightest metal**, and **osmium** is the **densest (heaviest) metal**.

Metal	Atomic Symbol	Density
Gold	Au	19.3
Silver	Ag	10.5
Mercury	Hg	13.6
Platinum	Pt	21.45

BCW Fact: **White gold** is an alloy of **gold** mixed with metals such as **nickel, palladium, or silver**.

Reactivity Series of Metals

- The **metal reactivity series** is a list of metals arranged in **decreasing order of activity**. Metals like **K, Na, Ca, Mg, and Al** are highly reactive and are **not found in nature** as free elements.
- Platinum** is one of the **least reactive metals** and shows excellent **corrosion resistance**, so it is considered a **noble metal**.
- Metals at the top of the series are **strong reducing agents** because they are **easily oxidised** and corrode readily.
- The **electropositive character** of metals decreases down the reactivity series.
- Metals placed **above hydrogen** in the series liberate **H₂ gas** when reacting with **dilute HCl or dilute H₂SO₄**.
- More reactive metals can **displace less reactive metals** from their salt solutions.
- Metals higher in the series require **more energy for extraction** from their ores.
- The **electron-donating tendency** of metals decreases down the reactivity series.

Metals	Reactivity
Potassium	Reacts with water
Sodium	
Lithium	
Barium	
Strontium	
Calcium	Reacts with acids
Magnesium	
Aluminium	
Manganese	
Zinc	
Chromium	
Iron	
Cadmium	
Cobalt	
Nickel	
Tin	Included for comparison
Lead	
Hydrogen	Highly unreactive
Antimony	
Bismuth	
Copper	
Mercury	
Silver	
Gold	
Platinum	

Although **hydrogen is not a metal**, it has been assigned a **place in the reactivity series of metals** because, like metals, hydrogen also has a tendency to lose electrons and forms a positive ion, H⁺. The metals that lose electrons less readily than hydrogen are placed below it, and the metals that lose electrons more readily than hydrogen are placed above it in the reactivity series of metals.

BCW Fact: Food cans are **coated with tin** and not with zinc because zinc is above tin in the reactivity series. It means Zinc is more reactive than tin and can react with food elements preserved in it.

Corrosion

- Corrosion** is the gradual deterioration of metals due to the formation of **oxides or salts** on their surfaces. Examples include **rusting of iron, tarnishing of silver, and green coating on copper and bronze**.
- In corrosion, metals undergo **oxidation** by losing electrons, usually in the presence of **oxygen and moisture**.
- Silver** turns black due to the formation of a coating of **silver sulphide**.
- Copper** reacts with **moist carbon dioxide and oxygen** in air to form a green coating of **basic copper carbonate**.
- Iron** exposed to moist air develops a **brown flaky coating called rust**.

Corrosion Prevention

- Application of paint, oil, or grease.
- Galvanization:** The process of **applying a zinc coating to iron or steel**, serving as a protective shield against rust.
- Electroplating:** The technique of utilizing electrical current to apply a thin layer of a corrosion-resistant metal, such as chromium or nickel, onto the surface of another metal for protection.
- Alloying:** Mixing **metals to create alloys** like stainless steel, which are more resistant to corrosion.

Non-Metals

Non-metals are elements that do not have the characteristics of metals. They are located on the right side of the periodic table. Some of the properties of non-metals are:

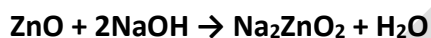
- Non-metals are **poor conductors of heat and electricity**. They are **neither malleable nor ductile**. They are **not shiny**.
- When dissolved in water, non-metals produce **acidic oxides**.
- **Iodine** is a non-metal but is lustrous.
- Non-metals have **low melting and boiling points**.
- They have **high electronegativity and high ionization energy**.

Amphoteric Oxides

- Those oxides that behave as both acidic and basic or that can react with acid, as well as base, are called **amphoteric oxides**.
- **Example:** Aluminium oxide (Al_2O_3), zinc oxide (ZnO), lead monoxide (PbO) and stannic oxide (SnO_2).
- As a basic oxide, ZnO reacts with hydrochloric acid forming, zinc chloride and water.



- As an acidic oxide, ZnO reacts with Sodium hydroxide (base) to form sodium zincate and water.



Thermite Reaction

- The process of **igniting aluminium and ferric oxide (Fe_2O_3)** is called the thermite reaction. This is an exothermic reaction.
- Iron obtained in this process is in the molten state. The thermite reaction is used for the **welding of rails (joining metals)**, a process known as welding.

Periodicity in Properties of Elements

Property	In Period (From left to Right)	In Groups (From top to bottom)
Atomic Size	Decreases	Increases
Atomic Radius	Decreases	Increases
Ionic Radius	Decreases	Increases
Ionization Energy/Potential	Increases	Decreases
Electro-negativity	Increases	Decreases
Electron Affinity	Increases	Decreases
Melting point	Increases	Decreases
Nature of Hydrides	Becomes acidic from alkaline	Increase in acidic nature
Nature of Oxides	Alkalinity decreases and acidity increases	Alkalinity Increases and acidity decreases
Metallic characteristics	Decreases	Increases
Oxidising nature	Increases	Decreases

BCW Facts:

- **Technetium (Tc)**, designated as the chemical element with atomic number 43, holds the distinction of being the **first element produced artificially**. Technetium is a silvery-gray

radioactive metal. It was successfully isolated by scientists Carlo Perrier and Emilio Segre in the year 1937.

- **Radium (Ra, atomic number 88)** was discovered in 1898 by Marie Curie and Pierre Curie, who managed to extract 1 mg of radium from ten tonnes of the uranium ore pitchblende (uranium oxide, U_3O_8).
- **Californium** is a member of the **actinoid series** in the periodic table. With an atomic number of 98, it is a **transuranium element**, meaning it has an atomic number greater than that of uranium (92).

10- Chemistry In Everyday Life

Drugs and medicines play a crucial role in modern healthcare by aiding in the prevention, diagnosis, and treatment of diseases. Key categories of medicinal drugs, including **antacids**, **neurologically active compounds**, and **antimicrobials**, help manage various health conditions affecting millions globally. Likewise, the **role of chemistry in food and fertilizers** is crucial for enhancing food security, improving nutrition, and promoting sustainable agriculture.

Drugs And Their Classification

- **Drugs** are chemicals of **low molecular mass (~100–500 u)** that interact with **macromolecular targets** to produce a biological response. When this response is **therapeutic and useful**, the drugs are called **medicines**.
- Pharmaceuticals generally interact with biological macromolecules such as **carbohydrates, proteins, lipids, and nucleic acids**, known as **target molecules**.
- Drugs that bind to **receptor sites** and block their normal action are called **antagonists**. They are useful when a biological message needs to be blocked.
- Conversely, there are other medications that imitate the natural messenger by activating the receptor; these are known as **agonists**. These are useful when there is a lack of a natural chemical messenger.

Antacids

- **Antacids** are drugs used to treat **excess acid production in the stomach**, which causes **acidity and pain**. They work by **neutralising stomach acid**.
- Until 1970, acidity was mainly treated with antacids such as **sodium hydrogen carbonate** or a mixture of **aluminium and magnesium hydroxides**.
- However, excessive use of **sodium hydrogen carbonate** can make the stomach too **alkaline**, leading to increased acid production.

Antihistamines:

- **Allergies** are hypersensitivity reactions caused by the release of the chemical **histamine** in the body.

- Histamine plays a significant role in various conditions, including **nasal congestion, mild asthma, cough, and pollen allergies**. It causes **contraction of smooth muscles** in the bronchi and gastrointestinal tract and relaxes blood vessel walls.
- **Antihistamines** are drugs that **block or reduce the effects of histamine** and are therefore called **anti-allergic drugs**.
- Examples of synthetic antihistamines include **brompheniramine (Dimetapp), terfenadine (Seldane), and diphenhydramine (Benadryl)**.

Neurologically Active Drugs

1. Tranquilizers

- **Tranquilizers** are neurologically active drugs used to treat **anxiety and related disorders**. They reduce anxiety by **slowing down the central nervous system** and are also called **anxiolytics or sedatives**. Examples include **benzodiazepines** such as **Valium** and **Xanax**.
- **Chlordiazepoxide** and **meprobamate** are relatively **mild tranquilizers** used to relieve tension.
- **Equanil** is used to control **depression and hypertension**.
- **Barbiturates** (derivatives of barbituric acid) such as **amytal, luminal, veronal, nembutal, and seconal** are important tranquilizers that act as **hypnotics (sleep-inducing agents)**.

2. Analgesics

- **Analgesics** relieve pain **without causing loss of consciousness or impairing the nervous system**.

Types of Analgesics:

1. **Non-narcotic (non-addictive): Aspirin and paracetamol.**
2. **Narcotic drugs: Morphine** and its homologues, which relieve pain and may induce sleep.

BCW Fact: Aspirin (Acetylsalicylic Acid) is widely recognized for its properties as an **analgesic, antipyretic, and anti-inflammatory agent**. Due to its ability to inhibit blood clotting, aspirin is used to prevent **heart attacks**. The natural compound **salicin**, which serves as the precursor to aspirin, is derived from the bark of the willow tree, belonging to the genus *Salix*.

Antimicrobials

- **Antimicrobials** destroy or inhibit the growth of **pathogenic microbes** such as bacteria, fungi, viruses, and parasites. Examples include **antibiotics, antiseptics, and disinfectants**.

1. Antibiotics

- **Antibiotics** are drugs used to treat **bacterial infections** and generally have **low toxicity for humans and animals**. They either kill bacteria or inhibit their growth.

Types of Antibiotics

- **Bactericidal:** Kill bacteria by disrupting vital processes. Examples: **Penicillin, Ofloxacin, Aminoglycosides**.
- **Bacteriostatic:** Inhibit bacterial growth without killing them directly. Examples: **Erythromycin, Chloramphenicol, Sulfadiazine, Tetracycline**.

BCW Fact: Penicillin was the first antibiotic to be discovered. In 1929, Alexander Fleming isolated penicillin from the mould **Penicillium notatum**.

2. Antiseptic

- Antiseptics are chemicals that **kill or inhibit the growth of microorganisms**. They are applied to **living tissues** such as wounds, cuts, ulcers, and infected skin. They are **not ingested like antibiotics**.

Commonly used antiseptic

- **Dettol**: mixture of *chloroxyleneol* and *terpineol*
- **Bithionol (bithional)**: added to soaps for antiseptic action.
- **Iodine**: powerful antiseptic; its **2–3% solution in an alcohol–water mixture = tincture of iodine**.
- **Iodoform**: used for wounds.
- **Boric acid** (dilute solution): weak antiseptic for eyes.

3. Disinfectants

- Disinfectants are used on **inanimate objects** like floors, instruments, and drainage systems.
- Same chemical may act as **antiseptic or disinfectant depending on concentration**.

Phenol: 0.2% solution → antiseptic | 1% solution → disinfectant

- **Chlorine (0.2–0.4 ppm in an aqueous solution)** and **sulphur dioxide (low concentration)** are disinfectants.
- Their **potent formulation** makes **disinfectants** unsuitable for **direct use on the human body**, as they can cause irritation or harm to living tissues.

Anti-fertility Drugs

- Anti-fertility drugs are chemical substances that **suppress the action of hormones that promote pregnancy**. Anti-fertility drugs are made up of **derivatives of synthetic progesterone** or a **combination of derivatives of estrogen and progesterone**.
- **Norethindrone** is an example of a synthetic progesterone derivative, most widely used as an **anti-fertility drug**. The estrogen derivative, which is used in combination with the progesterone derivative, is **ethynylestradiol (novestrol)**.

Chemical Fertilizers

Chemical Fertilizers are nutrients for plants that are produced on a commercial scale. They provide essential elements such as **nitrogen, phosphorus, and potassium**. Commonly used fertilizers include **nitrogenous, phosphatic, potassic, and complex fertilizers**.

1. Nitrogenous fertilizers

- **Urea**: Chemical formula $\text{CO}(\text{NH}_2)_2$. It is the **first organic compound synthesized in a laboratory**. It was synthesized by German chemist **Friedrich Wöhler (1828)** by heating **ammonium cyanate**. Urea is also known as **carbamide**. It is widely used in the **fertilizer industry, plastic industry, and pharmaceutical drugs**.
- Other commonly available nitrogenous fertilizers are **ammonium sulphate $[(\text{NH}_4)_2 \text{SO}_4]$** and **Calcium ammonium nitrate**, etc.

Neem-coated urea is an enhanced version of traditional urea fertilizer, featuring granules that are coated with **neem oil or extracts**. This innovation was developed to improve **nitrogen efficiency**. This coating acts as a **nitrification inhibitor**, ensuring a **gradual nitrogen release**, which aligns with crop needs and reduces **nutrient loss, leaching, and volatilization**. Additionally, its usage is **10% lower than standard urea**, leading to **cost savings and better environmental sustainability**.

2. Phosphatic fertilizers

- Primary source of phosphorus.
- **Examples**: single superphosphate, triple superphosphate, and dicalcium phosphate.

3. Potassic fertilizers

- Used as a source of potassium.
- **Examples:** potassium chloride or muriate of potash and potassium sulphate.
- 4. Mixed fertilisers (complex fertilisers)**
- Fertilizers that supply more than one essential macronutrient—Nitrogen (N), phosphorus (P) and Potassium (K) — for plant growth, are known as mixed fertilizers.
- **Examples:** di-ammonium phosphate (18:46:0), **nitro-phosphate** (20:20:0), **ammonium phosphate** (28:0:0), and **ammonium phosphate sulphate** (16:20:0).

For example, a 100-kilogram bag of di-ammonium phosphate, which consists of 18% nitrogen (N) and 46% phosphorus (P), includes 18 kilograms of nitrogen and 46 kilograms of phosphorus, with the remaining 36 kilograms comprising inert material.

Artificial Sweetener

Artificial sweeteners are **synthetic sugar substitutes** that provide **little or no calories** and are therefore called **non-nutritive sweeteners**. They are widely used by **diabetics** and people who wish to reduce sugar intake.

Saccharine

- Ortho-sulphobenzimide, commonly known as **saccharin**. First artificial sweetener, discovered in **1879**.
- About **550 times sweeter** than **cane sugar**.
- Not metabolized by the body → **zero calorific value**
- Commonly used by diabetics and people who wish to reduce sugar intake.

Aspartame:

- Methyl ester of a dipeptide formed from **aspartic acid and phenylalanine**.
- About **100 times sweeter** than **cane sugar**.
- Used in **cold foods and soft drinks**.
- **Unstable at cooking temperatures**.

Alitame

- About **2000 times sweeter** than **cane sugar**.
- More stable than aspartame and suitable for **cooking and baking**.
- Difficult to control sweetness level in foods.

Sucralose

- Sucralose, a substituted disaccharide, is a non-nutritive sweetener.
- Prepared by the **selective chlorination of sucrose**.
- About **600 times sweeter** than cane sugar.
- Sucralose also has **excellent stability in low-pH products**.
- Widely used in **foods, beverages, and pharmaceuticals** without adding calories.

Important Chemical Compounds

Blue Vitriol (Neela Thotha)

- Blue vitriol (Neela Thotha) is **copper sulphate pentahydrate** with the chemical formula **$\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$** . Copper sulphate is used as a **fungicide, herbicide, and algicide**. It is also used in **water purification and sterilization systems** because of its antibacterial properties.
- The number of water molecules attached to one formula unit of a compound is called **water of crystallization**. In blue vitriol, **5 water molecules** are attached to one formula unit. Blue vitriol has a water of crystallization value of 5.

Breath Analyzer

- A breath analyzer consists of a **mouthpiece** and two chambers containing liquid connected to a meter that detects a colour change. The chamber contains compounds such as **potassium dichromate, sulphuric acid, and silver nitrate** (catalyst).
- To use the Breath analyzer, the subject exhales through the mouthpiece into a test chamber filled with a reddish-orange solution of **potassium dichromate ($\text{K}_2\text{Cr}_2\text{O}_7$)**.
- Alcohol reacts with potassium dichromate, turning the solution **green**. The greater the amount of ethanol present, the greater the green colour produced. This process involves **oxidation and reduction reactions**.

Glass

- Glass is a **non-crystalline (amorphous), transparent or translucent solid** that is hard and brittle. It is usually formed by **rapid cooling (quenching)** of molten material. Some types, such as **volcanic glass**, occur naturally.
- Most manufactured glass is **silicate glass**, made mainly from **silica (SiO_2)**, the chief component of sand.
- Soda-lime glass**, containing about **70% silica**, accounts for nearly **90% of manufactured glass**.
- In glass, various **metallic oxide impurities** are added to give the glass a **typical colour**.
 - Chromium oxide** : Green colour
 - Manganese dioxide** : Violet colour
 - Cadmium sulphide** : Yellow colour
 - Cobalt oxide** : Deep blue colour

Sodium Azide

- Sodium azide (NaN_3)** is used in **airbags**. During a collision, sensors trigger its decomposition, releasing a large amount of **nitrogen gas (N_2)**.
- About **130 g of sodium azide produces nearly 67 litres of nitrogen gas**, which rapidly inflates the airbags and helps protect passengers.

BCW Facts:

- Antimalarial medicines** are used to treat **malaria**. **Quinine** and **chloroquine** are commonly used antimalarial drugs. Quinine, one of the earliest antimalarial drugs, was first obtained from the bark of the **Cinchona tree** and later synthesized in laboratories.
- Antipyretics** are drugs used to **reduce fever and lower body temperature**. Common examples are **aspirin, paracetamol, and phenacetin**.
- Triclosan** is a synthetic chemical with **antibacterial and antifungal properties**. It is commonly used in **antibacterial soaps, hand washes, gels, household cleaners, toothpaste, deodorants, and facial cleansers**.
- Gambusia fish (mosquitofish)** are used as a **biological control agent** to reduce mosquito populations by feeding on mosquito larvae in water bodies. This helps in controlling diseases such as **malaria and dengue**.

11- The Fuel

Core Concepts & Classifications

- **Fuel Definition:** Substances that undergo combustion (exothermic reactions) to release heat and light.
- **Calorific Value:** Heat energy produced by complete combustion of 1 kg of fuel. Unit: **kJ/kg**.
 - *Highest Calorific Value:* **Hydrogen gas (H_2)** at **150,000 kJ/kg**.

Ideal Fuel Characteristics:

- High calorific value.
- Moderate ignition temperature.
- Leaves no residue (ash) or toxic gas emissions after burning.
- Cheap, readily available, and easy to transport.

Classification by Source:

- **Fossil Fuels:** Coal, Petroleum, Natural Gas. (Formed from dead remains, depletable).
- **Renewable Fuels:** Biofuels (Ethanol, Biodiesel, Biogas), Green Hydrogen.

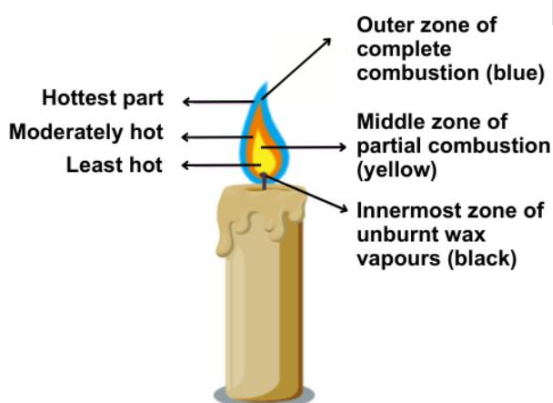
Classification by State: Solid (Coal, Coke), Liquid (Petrol, Kerosene), Gaseous (CNG, Biogas).

Combustion, Flame Zones & Fire Control

- **Ignition Temperature:** The lowest temperature at which a substance catches fire.

Zones of a Candle Flame:

- **Outermost Zone (Blue):** Complete combustion — hottest part. Used by goldsmiths to melt precious metals.
- **Middle Zone (Yellow):** Incomplete combustion — moderately hot and luminous.
- **Innermost Zone (Black):** Unburnt vapours — least hot part.

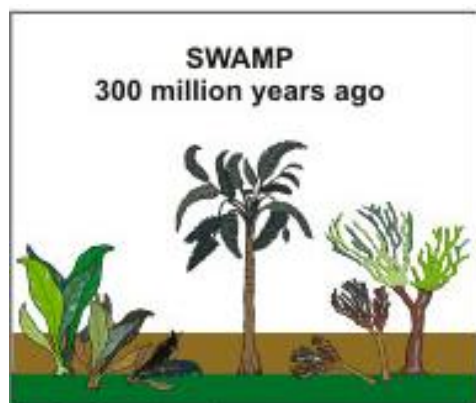


Fire Control Chemistry:

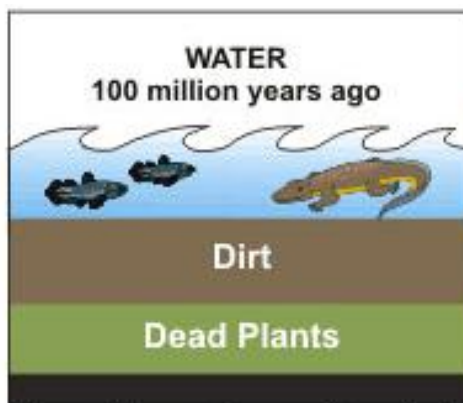
- **Oil Fires:** Water fails because water is heavier than oil; it sinks while oil continues to burn on top.
- **Best Extinguisher for Oil/Electrical Fires:** Carbon Dioxide (CO_2). Heavier than oxygen, it forms a blanket over the fire, cutting off the oxygen supply.
- **Soda-Acid Extinguisher:** Uses $NaHCO_3 + H_2SO_4$ to instantly generate CO_2 gas.

Coal: Formation, Types & Processing

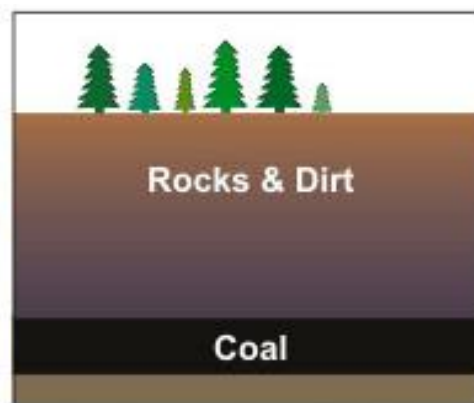
- Basics:** Combustible sedimentary rock primarily composed of carbon, along with variable amounts of other elements, mainly hydrogen, sulphur, oxygen and nitrogen. Coal is a type of fossil fuel, which is formed when dead plant matter decays into peat and is converted into coal from the heat and pressure of deep burial over millions of years.



Before the dinosaurs, many giant plants died in swamps.



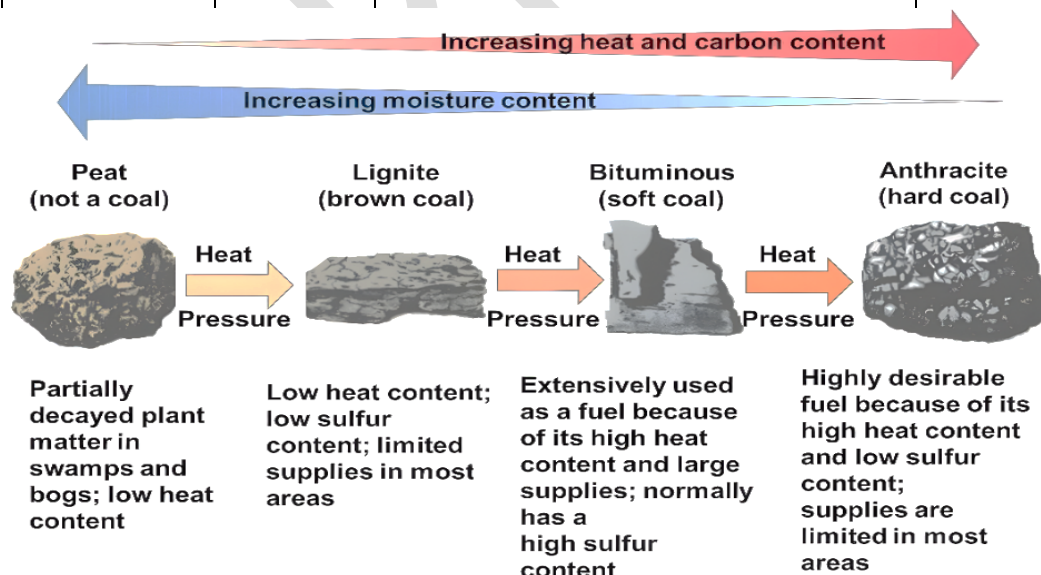
Over millions of years, the plants were buried under water and dirt.



Heat and pressure turned the dead plants into coal.

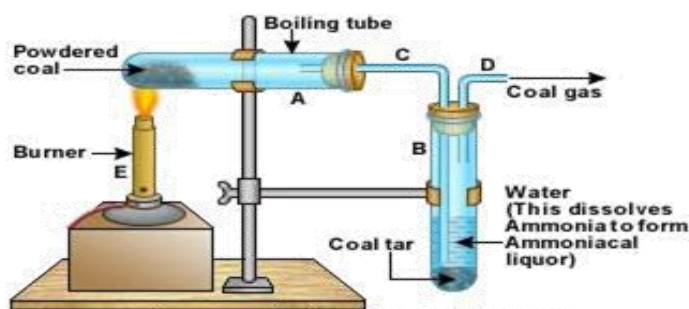
Types of Coal (Based on Carbon % & Quality):

Coal Type	Carbon %	Key Characteristics	Key Indian Reserve
Peat	~40%	Pre-coal stage, lowest quality, emits high CO ₂ /CO.	North-East India
Lignite (Brown)	40–60%	Low heat content; low sulfur.	Neyveli, Tamil Nadu
Bituminous (Soft)	60–80%	Most extensively used. High heat, high sulfur.	Jharkhand, Odisha
Anthracite (Hard)	80–90%	Best quality, lowest ash content.	Reasi, J&K



Destructive Distillation (heating coal in the absence of air) yields:

- **Coke:** ~Pure carbon. Reducing agent in blast furnaces to extract iron from Fe_2O_3 .
- **Coal Tar:** Thick liquid (~200 substances). Used for naphthalene balls, dyes. Now replaced by Bitumen for road metalling.
- **Coal Gas:** Industrial heat source. First used for London streetlights in 1810.



- **Emissions:** Burning coal releases CO_2 , CO, SO_2 , Nitrogen, Lead, Arsenic, and Mercury.

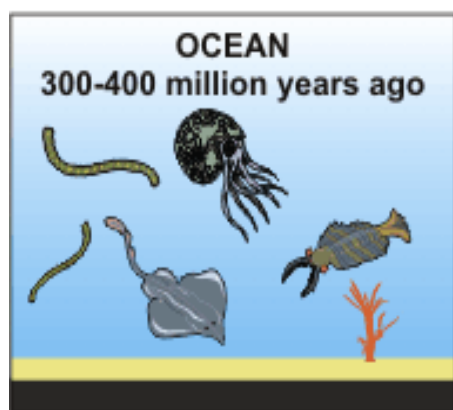
India's Coal Profile

- **Production & Ranking:** India crossed 1 Billion Tonnes (March 20, 2025). Ranked 5th globally in reserves, 2nd in consumption.
- **Energy Mix:** Coal constitutes 55% of India's total energy mix.
- **Dominant Type:** ~80% of Indian deposits are Bituminous (predominantly non-coking grade).
- **Indian Coal Ash:** Unusually high ash content (35–50%) compared to global averages.
- **Key Mines:**
 - **Oldest:** Raniganj, West Bengal (first mined in 1774).
 - **Largest:** Jharia, Jharkhand.
 - **Top Reserves:** Odisha and Jharkhand.

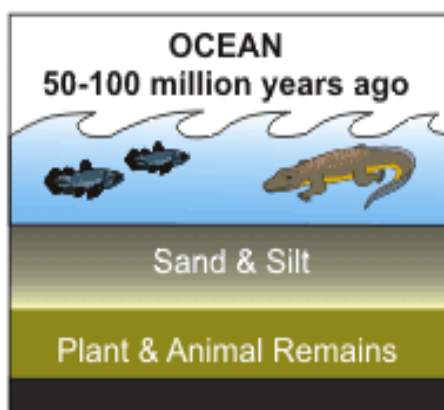
Petroleum ("Black Gold" / "Liquid Gold")

The word 'petroleum' comes from two words – '**Petra**' which means 'rock' and '**oleum**' which means 'oil'. It is the oil extracted from the rocks beneath the earth. Petroleum is a chemically **complex mixture of hydrocarbon compounds**, primarily consisting of hydrogen and carbon, along with trace amounts of nitrogen, oxygen, and sulphur as impurities. The liquid form of petroleum is referred to as crude oil.

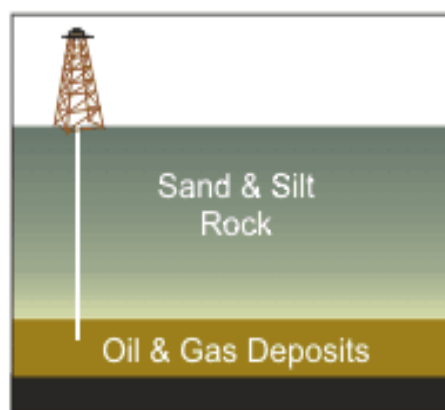
- **Formation:** Marine organisms (Planktons/Algae) → Kerogen → Hydrocarbons.
- **Extraction Stratification (top to bottom):** Natural Gas → Crude Oil → Water.



Tiny sea plants and animals died and were buried on the ocean floor. Over time, they were covered by layers of silt and sand.



Over millions of years, the remains were buried deeper and deeper. The enormous heat and pressure turned them into oil and gas.



Today, we drill down through layers of sand, silt, and rock to reach the rock formations that contain oil and gas deposits.

- **Historical Oil Wells:**
 - 1st globally: Pennsylvania, USA (1859).
 - **1st in India (2nd globally):** Makum, Assam (1867).
- **India's Oil Locations:** Assam, Gujarat, Mumbai High, Godavari/Krishna river basins.
- **Refining Process:** Fractional Distillation based on boiling points. Low BP gases rise to the top; heavy oils (Bitumen) settle at the bottom.

FRACTIONAL DISTILLATION CRUDE OIL



Petroleum By-Products & Synthetic Gases

- **LPG:** Composed of Butane, Propane, Ethane (no Methane). Ethyl mercaptan (C_2H_5SH) added for leak-detection odour.
- **Natural Gas / CNG / PNG:** Primary component is Methane. Clean fuel, no residue. Found in Tripura, Rajasthan, Maharashtra, Krishna-Godavari Delta.
- **Kerosene:** Contains no oxygen. Used as aviation fuel in jet engines.
- **Paraffin Wax:** Used for Vaseline, creams, and candles.
- **Bitumen:** Used for paints and road paving/metalling.

- **Petcoke (Petroleum Coke):** Carbon-rich solid from cracking. Used in cement, lime kilns, and aluminium anodes.
- **Furnace Oil:** Heavy residual oil. Combustion releases SO_2 , NO_x , and particulate matter.

Important Industrial Gas Mixtures:

- **Water Gas:** $\text{CO} + \text{H}_2$
- **Producer Gas:** $\text{CO} + \text{N}_2$ (lowest calorific value among gaseous fuels)
- **Synthesis Gas (Syngas):** $\text{CO} + \text{CO}_2 + \text{H}_2$ — used to produce synthetic petroleum

Fuel Quality Indicators & Additives

- **Octane Number:** Measures petrol/gasoline quality. Higher number → better resistance to engine knocking.
- **Cetane Number:** Measures diesel quality. Higher number → faster/better ignition efficiency.
- **Tetraethyl Lead (TEL):** Former anti-knock petrol additive. Completely phased out due to severe neurological damage from lead exposure.
- **Ethylene Glycol:** Used as an anti-freeze for cooling car engines.

Renewable Fuels & Biofuels

- **Biodiesel:** Produced from vegetable oils, jatropha, or algae via trans-esterification. Higher H/O ratio, lower carbon than fossil fuels.
- **Bioethanol:** Produced via fermentation of sugarcane, corn, and molasses.
- **Ethanol Blending Target (E20):** India advanced to the 2025–26 supply year (originally 2030). Ethanol-blended petrol is called Gasohol.
- **Biogas / Gobar Gas:** Produced via anaerobic decomposition of biomass/cow dung. Primary components: CH_4 , CO_2 , and trace H_2S .

Hydrogen Fuel Types

- **Green Hydrogen:** Produced via electrolysis of water using renewable energy (wind/solar/hydro). Zero carbon emissions.
- **Grey Hydrogen:** Produced via Steam Methane Reforming (methane + high-temperature steam). Emits CO_2 .

Key Government Initiatives & Targets

Panchamrit Targets (by 2030):

- Expand non-fossil energy capacity to 500 GW.
- Meet 50% of energy needs from renewable sources.
- Decrease carbon intensity of the economy by 45%.
- Reduce projected carbon emissions by 1 billion tonnes.
- **Net-Zero Emissions Target:** 2070.
- **National Green Hydrogen Mission:** Target of 5 million tonnes/annum production by 2030.
- **PM Ujjwala Yojana (2016):** Provides LPG connections to underprivileged households for clean cooking.
- **ISA (International Solar Alliance):** Established by India & France in 2015 at COP21 (Paris Climate Conference).
- **PCRA (Petroleum Conservation Research Association):** Reconstituted in 1978 under Ministry of Petroleum & Natural Gas to promote energy efficiency.

12 – Pollution

The Atmosphere: Composition & Layers

The layer of air present around the earth is called the atmosphere

Composition

- **Nitrogen:** 78% | **Oxygen:** 21% | **Argon:** 0.93% | **CO₂:** ~0.03% | **Others:** 0.4%

Atmospheric Layers

- **Troposphere (0–13 km):** Where weather occurs. CO₂ and 'bad' ozone act as pollutants here.
- **Stratosphere (13–50 km):** Contains the protective '**good**' **Ozone Layer**. Ideal for aircraft.
- **Mesosphere (50–80 km):** Most meteors burn up here.
- **Thermosphere / Ionosphere (80–600 km):** Reflects radio waves back to Earth. Houses the **ISS**.
- **Exosphere:** Merges into outer space.

Air Pollution & Pollutant Classifications

- **Primary Pollutants:** Emitted directly into the atmosphere (e.g., CO, CO₂, SO₂, NO_x, Particulate Matter).
- **Secondary Pollutants:** Produced via chemical reactions of primary pollutants in air (e.g., **Ozone**, **PAN** [Peroxyacetyl nitrate], Sulphuric/Nitric acid, Smog).

★ **Key Fact:** *Lichens* are the ultimate biological indicator of **Sulphur Dioxide (SO₂)** pollution. Their disappearance from a forest signals severe SO₂ pollution.

Major Gaseous Air Pollutants

- **Carbon Monoxide (CO):** Combines with haemoglobin to form **Carboxyhaemoglobin**, reducing oxygen-carrying capacity. Fatal at **750 ppm**.
- **Oxides of Nitrogen (NO_x):** Nitrous Oxide (N₂O) = "**Laughing Gas**" — both an ozone-depleting substance and a greenhouse gas.
- **Sulphur Dioxide (SO₂):** Causes rigidity in flower buds, making them drop off plants.
- **Methane (CH₄ / Marsh Gas):** Emitted from swamps, **paddy fields**, and the gut of **ruminant animals**.
- **3,4-Benzopyrene:** Released by incomplete burning; a carcinogen linked to lung cancer.

Particulate Matter (PM)

Particulate Matter

- **PM_{2.5} (≤2.5 μm):** Penetrates deep into lungs and bloodstream; causes asthma and heart attacks.
- **PM₁₀ (≤10 μm):** Trapped in the upper respiratory tract.
- **Specific Toxins:** **Lead** (brain/respiratory damage); **Silica / Arsenic / Asbestos** (carcinogens).

Photochemical Smog (Los Angeles Smog)

- Formed in **warm, dry, sunny** climates via sunlight reacting with unsaturated hydrocarbons and NO_x.
- **Constituents:** Ozone, Nitric Oxide, Acrolein, Formaldehyde, PAN.
- **Nature:** Oxidizing smog. Causes severe eye irritation; degrades rubber.

Classical Smog (London Smog)

- Formed in **cool, humid** climates.
- **Constituents:** Smoke, fog, and Sulphur Dioxide (SO_2).
- **Nature:** Reducing smog.

Acid Rain

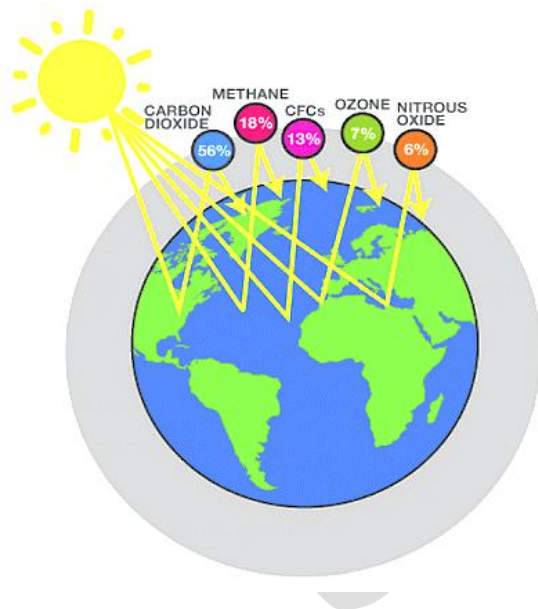
- **Definition:** Rainwater with pH **below 5.6**.
- **Primary Culprits:** SO_2 and NO_x .
- **Chemical Products:** H_2SO_4 (Sulphuric Acid) and HNO_3 (Nitric Acid).
- **Key Fact:** H_2SO_4 accounts for **60–70%** of rain's acidity.
- **Impact:** Damages stone buildings — yellowing of the **Taj Mahal** ('Stone Cancer') is a classic example.

Global Warming & Greenhouse Effect

- **Greenhouse Effect:** Gases trap **longwave (infrared) radiation** emitted from Earth's surface, preventing it from escaping into space.

Major Greenhouse Gases (by Contribution %)

- CO_2 (56%) > CH_4 (18%) > CFCs (13%) > Ozone (7%) > N_2O (6%)
- **Global Dimming:** Decline in sunlight reaching Earth due to sulphate aerosols. Cools the planet but disrupts the water cycle.



Ozone Depletion

- **Measurement:** Measured in **Dobson Units (DU)**. Depletion = dropping below **220 DU**.
- **Montreal Protocol:** Adopted **16th September 1987** (effective Jan 1, 1989). Targets CFCs and Halons.
- **World Ozone Day:** **September 16th**.
- **Note:** Ammonia (NH_3), CO, and CO_2 do not destroy ozone.

Water Pollution, BOD & Biomagnification

- **BOD (Biochemical Oxygen Demand):** Oxygen required by bacteria to decompose organic matter. **Clean:** <5 ppm | **Highly Polluted:** ≥17 ppm
- **Eutrophication:** Nutrient enrichment (N/P) causing algal blooms, oxygen depletion, and death of aquatic life.
- **Biomagnification:** Toxic substances (e.g., **DDT**) concentrate up the food chain. Apex predators suffer most (eggshell thinning).
- **'Terror of Bengal':** *Eichhornia crassipes* (Water Hyacinth) — a highly invasive aquatic weed.

Toxins & Associated Diseases

Pollutant	Source / Accumulation	Disease / Effect
Nitrates	Agricultural runoff	Methemoglobinemia ('Blue Baby Syndrome') — infant skin turns blue due to lack of oxygen.
Mercury	Industrial wastewater	Minamata Disease (severe neurological disorder).
Cadmium	Industrial waste	Itai-Itai ('Ouch-Ouch') Disease (weak, painful bones).
Arsenic	Groundwater (esp. Ganges Delta); accumulates in nails/hair.	Black Foot Disease
Fluoride	Groundwater	Skeletal Fluorosis and Knock-Knee Syndrome (legs bend outwards).

Noise Pollution Limits

Sound intensity is measured in **Decibels (dB)**. Chronic exposure above **80 dB** is hazardous. Permissible limits:

- **Residential Areas:** 55 dB (Day) / 45 dB (Night)
- **Silence Zones (Hospitals/Schools):** 50 dB (Day) / 40 dB (Night)
- **Industrial Areas:** 75 dB (Day) / 70 dB (Night)

Key Environmental Disasters & Acts

Bhopal Gas Tragedy

- Date: **December 2–3, 1984**
- Cause: Leak of highly toxic **Methyl Isocyanate (MIC)** from the Union Carbide plant.

DDT (Dichloro – diphenyl - trichloroethane): Used aggressively during WWII for malaria. Banned for agricultural use in India in **1989** due to extreme environmental toxicity.

Green Chemistry

- Coined by **Paul T. Anastas**. Focuses on designing processes that generate minimal pollution.

Chemistry of Firecrackers (Flame Colors):

- **Red:** Strontium salts (Strontium carbonate).
- **Green:** Barium compounds (Barium chloride/nitrate).
- **Blue:** Copper salts.
- **Yellow:** Sodium salts (Sodium nitrate/oxalate).
- **Orange:** Calcium salts.
- **Silver/White:** Aluminium, Magnesium, or Titanium.

Indian Environmental Acts

Act	Year
Wildlife (Protection) Act	1972
Water (Prevention and Control of Pollution) Act	1974
Forest (Conservation) Act	1980
Air (Prevention and Control of Pollution) Act	1981
Environment (Protection) Act (enacted after Bhopal disaster)	1986